

Prototype-driven expert voting for pseudo-label selection with dual-balanced loss for imbalanced medical image classification

Sulei Wang^{a,1}, Shihao Sheng^{b,c,1}, Tiegong Wang^{d,1}, Xudong Liang^a, Xiao Chen^{b,c}, Hao Shen^{b,c}, Haoyang Zhang^a, Jiang Xie^{a,*}, Guangchao Wang^{b,c,**}, Jiacan Su^{b,c,e,**}

^a School of Computer Engineering and Science, Shanghai University, Shanghai, 200444, China

^b Department of Orthopaedics, Xinhua Hospital Affiliated to Shanghai JiaoTong University School of Medicine, Shanghai, 200092, China

^c Trauma Orthopaedics Center, Xinhua Hospital Affiliated to Shanghai JiaoTong University School of Medicine, Shanghai, 200092, China

^d Department of Radiology, Changhai Hospital, Naval Medical University, No. 168 Changhai Road, Yangpu District, Shanghai, 200433, China

^e MedEng-X Institutes, Shanghai University, Shanghai, 200444, China

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ABSTRACT

Medical image datasets inherently suffer from class imbalance, label scarcity, and intra-class heterogeneity. Under semi-supervised learning, pseudo-label selection remains a central research focus. Class prototypes constructed in the latent space have been studied for pseudo-label selection, particularly because sub-class prototypes can effectively represent within-class heterogeneity in medical imaging datasets. However, since sub-class prototypes are constructed using a relatively small number of samples, they may lead to local optima during prototype updating and usage. In addition, loss weight rebalancing is commonly used to mitigate attention bias in model. Existing methods tend to focus exclusively on improving the minority class, without taking into account ambiguous samples, which are often difficult to learn and contain valuable information. Here, a prototype-driven framework PDMatch is proposed, incorporating Prototype-Driven Expert Voting (PDEV), Majority-class Pseudo-label Under-Sampling (MPUS) and Dual-Balanced Loss (DBL). First, an Expert Prototype Set is constructed, and hierarchical updating is proposed to sufficiently capture the feature distributions. PDEV then generates and filters pseudo-labels through prototype-driven voting, where multiple candidate sub-class prototypes are jointly considered. Second, MPUS offers a concise post-processing strategy that reduces imbalance in the pseudo-label distribution. Finally, Dual-Balanced Loss rebalances loss weights to jointly account for class distribution and sample difficulty. Comprehensive experiments are conducted on three datasets with distinct imbalance structures, demonstrating that our model addresses inherent data challenges and consistently outperforms state-of-the-art methods across multiple metrics. Additionally, we compare our model with clinical expert assessments to underscore its practical value in real-world applications. Source code is available at <https://github.com/wang-su-lei/PDMatch>.

1. Introduction

Medical image datasets inherently exhibit imbalanced distributions, posing significant challenges for classification tasks. The imbalanced distribution is predominantly composed of majority classes, including normal disease cases, with only a small fraction representing minority classes consisting of rare diseases, as shown in Fig. 1. This severe class imbalance biases models toward the majority classes,

negatively impacting the accurate identification of minority classes [1]. Furthermore, expert-labeled data is difficult to obtain as it requires consensus among multiple clinical specialists. The scarcity of labeled samples restricts effective learning of minority classes, thereby exacerbating model bias [2,3].

Semi-supervised learning is widely used to address the lack of labeled samples by effectively exploiting information from unlabeled samples [4]. The mainstream methods can be categorized into consistency

* Corresponding author.

** Corresponding authors at: Department of Orthopaedics, Xinhua Hospital Affiliated to Shanghai JiaoTong University School of Medicine, Shanghai, 200092, China.
Email addresses: wang_su_lei@shu.edu.cn (S. Wang), doctorshengshihao@163.com (S. Sheng), wangtiegongwtg9@126.com (T. Wang), xudongliang@shu.edu.cn (X. Liang), sirchenxiao@126.com (X. Chen), shenhao@xinhua.com.cn (H. Shen), zhy-jsj-0105@shu.edu.cn (H. Zhang), jiangx@shu.edu.cn (J. Xie), wgc@shsmu.edu.cn (G. Wang), drsujacan@163.com (J. Su).

¹ These authors contributed equally to this work.

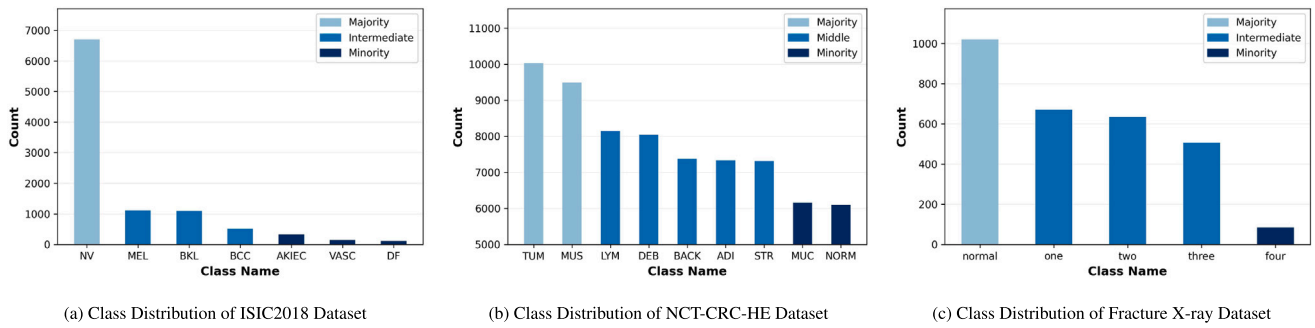


Fig. 1. Class distributions of the datasets, including (a) ISIC2018 (b) NCT-CRC-HE and (c) Fracture X-ray.

regularization [5–8] and pseudo-labeling [9–11]. Due to the inherent scarcity of labeled data in medical image datasets, consistency regularization is prone to overfitting, causing the model to degrade into a form of self-supervised learning [12]. Pseudo-labeling increases the labeling rate by selecting pseudo-labels, offering a solution to this critical challenge. As the field progresses, the xmatch paradigm has become widely adopted [13–15]. In this paradigm, unlabeled images undergo both weak and strong augmentations, where high-confidence predictions from the weakly augmented branch guide the unlabeled samples in the strongly augmented branch, alleviating the shortage of labeled data.

Numerous pseudo-labeling methods grounded in the xmatch paradigm have been explored, which can be classified into two types, namely probability-based and non-probability-based methods. Probability-based methods use predefined thresholds with the classifier's predicted probabilities as the selection criterion to identify high-confidence pseudo-label samples. In contrast, non-probability-based methods explore alternative metrics for pseudo-label selection, aiming to avoid classifier-induced bias and to uncover more informative samples.

Probability-based methods select pseudo-labels using predefined probability thresholds, including fixed and dynamic thresholds. The FixMatch [13] is a traditional framework with a fixed uniform threshold. As is well known, the majority class tends to have higher predicted probabilities, while minority classes rarely reach similar levels. Consequently, the fixed uniform threshold used in FixMatch [13] fails to effectively balance the selection of samples from both majority and minority classes. Recent studies [16,17] have proposed class-wise thresholding strategies that adaptively determine a separate dynamic threshold for each class, allowing low-confidence minority-class samples to be included in training and helping the model learn from more minority-class data. However, many high-confidence samples above the threshold may still carry incorrect pseudo-labels, especially in the early stages of training [18]. Continually introducing these incorrect yet high-confidence samples amplifies prediction bias, especially in imbalanced datasets where minority-class samples are often incorrectly classified as majority.

Recently, several methods [15,19] have explored non-probability-based selection strategies to identify more informative unlabeled samples from alternative perspectives. Class prototypes constructed in the latent space have been studied for pseudo-label selection, particularly because sub-class prototypes can effectively represent within-class heterogeneity in medical imaging datasets. In terms of class-prototype updating, compared with the batch-level weighted EMA updates [20–22] of mean class prototypes or the epoch-level average updates [23], the sub-class prototype set requires an appropriate sub-cluster partition due to the absence of explicit sub-cluster labels. Existing methods [24] do not sufficiently account for global dynamism in sub-cluster partitioning and rely on an early or random initialization of sub-class prototypes. Even with batch-level EMA updates, the sub-class prototypes initialized at early stages accumulate global partitioning biases, which may lead to local optima during training. In terms of prototype-based pseudo-label

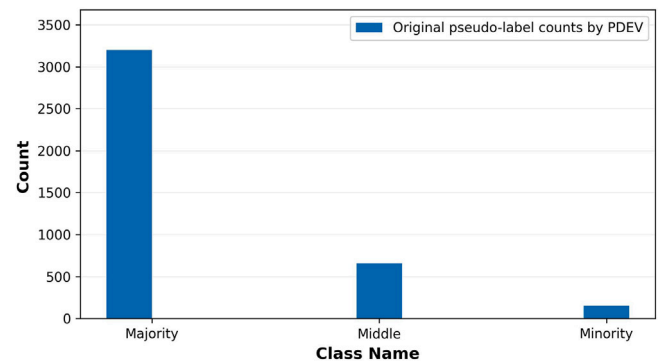


Fig. 2. The imbalanced distribution of pseudo-label quantities on ISIC2018 with 20% labeled data.

selection, common methods [21,22] select the nearest single mean class prototype as the pseudo-label, which ignores the intra-class feature distribution. Subsequent work [23] has represented complex feature distributions by clustering and computing the class distribution within each cluster. However, each sub-class prototype is constructed from a relatively small number of feature sub-clusters and cannot represent the class-level feature distribution. Therefore, sub-class prototypes used for pseudo-label generation are prone to local optima. Overall, although sub-class prototypes can capture fine-grained within-class feature distributions, they may also lead to local optima during prototype updating and usage.

Most pseudo-labeling methods directly use the selected pseudo-label subset for model training iterations. Further visualization experiments, as shown in Fig. 2, demonstrate that the pseudo-label subset also exhibits a significant class imbalance. Most methods ignore this imbalance in the selected pseudo-labeled samples. Under-sampling is an effective post-processing strategy often used in fully supervised frameworks [25,26] to handle class imbalance. However, in semi-supervised frameworks, due to the scarcity of labeled data, few methods have been proposed to perform under-sampling on the selected pseudo-labels. A portion of selected majority-class pseudo-label samples offers limited learning value and further amplifies prediction bias.

Besides addressing data imbalance, the model's attention also exhibits bias, which requires rebalancing through loss weighting. Balanced loss functions play a critical role in mitigating class imbalance in medical image analysis [27,28]. The standard cross-entropy loss assigns equal weight to all classes, which limits its effectiveness on imbalanced datasets. In the domain of class rebalancing, some studies [29,30] have adopted inverse class-frequency weighting, a static loss scheme that assigns fixed weights inversely proportional to the number of samples in each class. Recent studies [31,32] on balanced loss functions have employed novel and diverse metric strategies to enhance model robustness.

Moreover, on balanced datasets, certain methods [33–35] adjust weights based on model probabilities to emphasize ambiguous examples. Most existing methods have made significant progress. However, on imbalanced datasets, the majority of these methods focus exclusively on improving minority-class performance and do not consider ambiguous samples, which are often located near the decision boundary, such as those with low predicted probabilities or at the threshold of decision criteria. These samples are typically difficult to learn and contain valuable information.

Existing methods have substantially advanced semi-supervised learning for class-imbalanced classification tasks. Nevertheless, several further challenges persist. First, under the complex datasets of sparse labeled data, class imbalance, and intra-class heterogeneity, the use of subcluster-level prototypes for pseudo-label evaluation remains insufficiently explored, and this process imposes greater demands on the stability and responsiveness of prototype structures. Second, common pseudo-labeling methods in semi-supervised learning overlook the under-sampling post-processing of selected pseudo-labels, which exacerbates bias due to the inherent imbalance in the pseudo-label data distribution. Third, most loss weighting methods are designed to improve minority-class performance, yet ambiguous samples may not receive sufficient attention. Further exploration is needed to develop loss functions that can effectively employ fusion strategies to simultaneously take into account class distribution at the class level and learning difficulty at the sample level.

To address these issues, a prototype-driven framework PDMatch is proposed, incorporating Prototype-Driven Expert Voting (PDEV), Majority-class Pseudo-label Under-Sampling (MPUS), and Dual-Balanced Loss (DBL). The Prototypes Hierarchical Updating strategy has been designed for the prototype-based modules, in which batch-level and epoch-level updates alternate to ensure stable and timely construction of multi-cluster prototypes. PDEV has been proposed for pseudo-label generation and selection, where voting is performed within a kNN candidate set on the constructed fine-grained sub-class prototype set, increasing global information and improving robustness. MPUS is proposed on the subset filtered by PDEV and further filters majority-class unlabeled data by focusing on more informative instances. This undersampling strategy prevents low-information majority-class samples from dominating training and refocuses learning on rare and ambiguous instances. Moreover, to jointly account for class distribution and learning difficulty, DBL is proposed, which computes both aspects simultaneously and integrates them through an extremum-based fusion strategy.

The main contributions of this work are summarized as follows:

- Prototype-Driven pseudo-label selection is proposed, where global information is taken into account during the updating and utilization of sub-class prototypes to enhance robustness. PDEV performs voting using an Expert Prototype Set, which is constructed through Hierarchical Updating. This design provides a novel framework for prototype-driven pseudo-labeling under complex feature distributions.
- A Majority-class Pseudo-label Under-Sampling (MPUS) strategy is designed that selects more informative majority-class samples. This approach mitigates the impact of low-information majority-class unlabeled data, which would otherwise exacerbate bias under class imbalance.
- The sample-level Dual-Balanced Loss (DBL) is proposed, which jointly considers both the difficulty of pseudo-labeled samples and their class distribution. DBL emphasizes minority classes across classes and prioritizes ambiguous samples within each class.
- Extensive experiments are conducted on three datasets with distinct imbalance structures, demonstrating that our model overcomes existing data challenges and achieves higher performance than state-of-the-art methods. Moreover, the proposed model is

further evaluated against clinical expert judgments at varying levels of expertise to highlight its practical value.

2. Related work

2.1. Prototype learning

Prototype learning is widely used in fully supervised frameworks with imbalanced data distribution to construct representative class features [36]. A common approach maintains a class prototype for each category by using the mean feature [37–39]. PGPL [21] accumulates historical pseudo-label counts for each class to balance the sampling size and selects the top-k samples nearest to the class prototype based on the distance. Protocon [23] applies K-means to partition features into sub-clusters, estimates cluster-level class probabilities from labeled samples, and assigns these as soft pseudo labels to unlabeled samples within the same cluster. ProMatch [22] introduces two complementary prototypes for each class, one derived from feature embeddings and the other from model predictions. It requires the model's predicted class to match the class inferred from similarities to both prototypes. PALM [24] employs a multi-prototype structure for each class to support out-of-distribution detection. The method applies the Sinkhorn-Knopp algorithm to balance the soft assignment of samples across intra-class subclusters while preserving feature similarity. Prototypes have been adopted to capture finer-grained feature distributions, whereas the relatively small number of within-cluster samples poses a potential risk of local optima during updating and usage, which is worth deeper investigation.

2.2. Balanced weighted loss

Considering the variations in class distribution and learning difficulty in imbalanced medical image datasets, class-level weighted loss has become a widely adopted approach, with cross-entropy as its base loss function [1]. From the perspective of learning difficulty, Focal loss [33] first introduced predicted probabilities, enhancing the focus on ambiguous samples with lower probabilities. From the perspective of class distribution, CDMAD [35] addresses prediction bias at the probability level by incorporating awareness of class-distribution mismatch. CLD [40] computes the ratio of each class to the largest class as quantity-sensitive weights.

These methods individually calculate weights based on predicted probabilities, class distribution, or confidence, each showing distinct improvements. However, few loss functions comprehensively consider both factors. While class-balanced weighting formulas can mitigate class imbalance, they may overlook ambiguous samples. This issue requires further research to develop loss functions that simultaneously account for class distribution at the class level and learning difficulty at the sample level.

2.3. XMatch paradigm

The XMatch paradigm has been widely adopted for pseudo-label selection in semi-supervised learning, with FixMatch [13] representing its canonical form and stimulating extensive progress in the field [15,16,22]. The framework relies on a single encoder in which supervision is constructed through a single-model single-path mechanism driven by differently augmented views of the same image. For labeled samples, the ground-truth annotations are used to supervise both the weakly and strongly augmented images. For unlabeled samples, pseudo-labels are obtained from weakly augmented inputs, as the minimal perturbation in these views yields more stable and reliable predictions. These high-quality pseudo-labels are subsequently used to supervise their strongly augmented images, ensuring consistency even under substantial perturbations. To maintain pseudo-label fidelity, pseudo-labels obtained from strongly augmented images are rarely used to supervise weakly augmented ones [13]. Compared with the dual-model mutual-supervision design used in CPS [11], the single-model single-path structure of the XMatch paradigm is more suitable for medical imaging datasets, where

data volume is often limited. This design lowers the reliance on extensive data and computational resources, leading to a more efficient use of available labels and improved labeling performance.

3. Methods

3.1. Framework

As illustrated in Fig. 3, a prototype-driven framework named PDMatch is proposed, following the xmatch paradigm. PDMatch incorporates Prototype-Driven Expert Voting (PDEV), Majority-class Pseudo-label Under-Sampling (MPUS) and Dual-Balanced Loss (DBL). The main contributions focus on Prototype-Driven pseudo-label selection and finer-grained assignment of loss weights. In terms of Prototype-Driven pseudo-label selection, PDEV and MPUS are proposed. The Prototypes Hierarchical Updating strategy has been designed for the prototype-based module, in which batch-level and epoch-level updates alternate to ensure stable and timely construction of multi-cluster prototypes. PDEV evaluates the relations between unlabeled samples and the evolving class prototypes to perform voting. The voting results generate pseudo-labels and filter the high-confidence samples. MPUS builds upon the filtering results of PDEV and further discards less informative majority-class pseudo-labels to reduce class distribution imbalance. In terms of finer-grained assignment of loss weights, DBL jointly considers class distribution and sample difficulty through an extremum-based fusion strategy. For sample difficulty, higher weights are allocated to ambiguous samples. For class distribution, higher weights are allocated to less frequently occurring classes.

3.2. Prototype-driven for pseudo-label selection

3.2.1. Prototypes construction and hierarchical updating

Multiple sub-cluster prototypes are constructed for each class and treated as expert representations of intra-class characteristics to capture more detailed structures within such heterogeneous classes. To balance

global structural information with continuously updated local features, the Prototypes Hierarchical Updating (PHU) strategy is proposed, in which batch-level and epoch-level updates alternate to ensure stable and timely construction of multi-cluster prototypes. Specifically, at the beginning of each epoch, the expert prototypes are initialized using the global labeled features from the preceding epoch. During training iterations, the prototypes are updated with the latest mini-batch features. After the epoch concludes, the labeled features collected within that epoch are used for a new round of global initialization.

Considering the imbalanced class distribution, the Effective Number of Samples [41] is computed to estimate the relative contribution of each class, with a normalization applied with respect to the minimum class size.

$$E_c = \max \left\{ \left[\frac{1 - \beta^n}{1 - \beta^{n_{\min}}} \right], E_{\min} \right\}, \quad E_{\min} = \left[\frac{1 + C}{2} \right] \quad (1)$$

where n is the class sample count, $n_{\min}$ is the number of samples in the minority class, and C denotes the number of classes. $E_{\min}$ is the minimum number of class prototypes that prevents minority-class E_c from degenerating into a single prototype and grows slowly with the number of classes. β controls how fast E_c grows and is set to 0.999 following [41].

The epoch-level updates are performed before each epoch by applying k-means clustering to construct the initial expert prototypes. Specifically, for each class $c \in \{1, 2, \dots, C\}$, the encoder extracts features Z_c from its labeled samples. Then, the features Z_c are subjected to unsupervised k-means clustering, which partitions them into subclusters. E_c denotes the number of class prototypes for class c , and we use E_c as the k-value when performing k-means on Z_c . The mean feature of each subcluster is defined as its expert prototypes EP_c .

$$\{\mathcal{M}_{c,e}\}_{e=1}^{E_c} = \text{KMeans} \left(\underbrace{Z_c}_{\text{features}}, \underbrace{E_c}_{\text{cluster count}} \right) \quad (2)$$

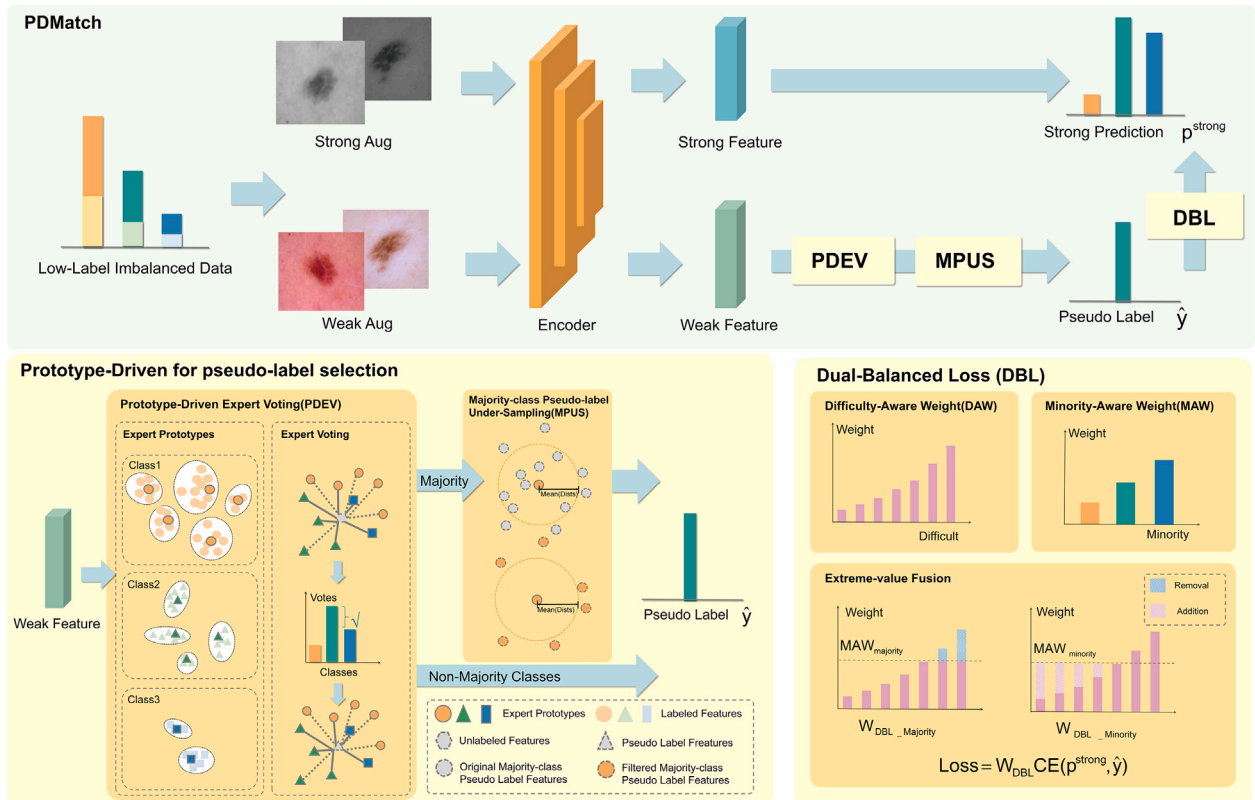


Fig. 3. The visualization of PDMatch incorporates Prototype-Driven Expert Voting, Majority-class Pseudo-label Under-Sampling, and Dual-Balanced Loss.

$$EP_c = \left\{ \mathbf{ep}_{c,e} = \frac{1}{|\mathcal{M}_{c,e}|} \sum_{z \in \mathcal{M}_{c,e}} z \mid e = 1, 2, \dots, E_c \right\} \quad (3)$$

where $\mathcal{M}_{c,e} \in \{\mathcal{M}_{c,1}, \dots, \mathcal{M}_{c,E_c}\}$ denotes the set of features belonging to the e -th subcluster of class c . $\mathbf{ep}_{c,e}$ denotes the expert prototype associated with the e -th subcluster of class c , and EP_c is the set of all expert prototypes belonging to class c .

The batch-level updates are performed to incorporate the latest learned feature into the expert prototypes. Specifically, for each labeled feature z_j^c in a mini-batch, the module locates the most similar same-class expert prototype $\mathbf{ep}_{c,i^*}$ and updates it via Exponential Moving Average (EMA).

$$\mathbf{ep}_{c,i^*} = \arg \max_{\mathbf{ep}_{c,i} \in EP_c} \frac{\langle z_j^c, \mathbf{ep}_{c,i} \rangle}{\|z_j^c\| \|\mathbf{ep}_{c,i}\|} \quad (4)$$

$$\mathbf{ep}_{c,i^*}^{batch} \leftarrow (1 - m_p) \mathbf{ep}_{c,i^*} + m_p z_j^c \quad (5)$$

where z_j^c denotes the feature of a labeled sample from class c in the mini-batch. EP_c is the set of all expert prototypes belonging to class c , as shown in Eq. (3). $\mathbf{ep}_{c,i^*}$ is the expert prototype with the highest similarity and $m_p = 0.9$ is a hyperparameter that controls the update ratio. The EMA update is applied sequentially to individual samples. Its convex-combination form performs long-term weighted averaging that gradually shifts the prototype toward the full data distribution. The specific order follows the model's reading sequence within each batch. Considering that it takes a convex-combination form, the process is insensitive to sample order.

In summary, the Prototypes Hierarchical Updating (PHU) strategy integrates global information with real-time local features, effectively mitigating the adverse influence of outdated early-stage representations on subsequent pseudo-label generation and selection. This enables the construction of expert prototypes that are both comprehensive and accurate.

3.2.2. Prototype-driven expert voting

Inspired by expert systems, a pseudo-label generation and selection strategy based on voting over expert prototypes is proposed, which exploits fine-grained features while additionally incorporating more global feature information. Specifically, for unlabeled sample features in each mini-batch, the Euclidean distance formula is employed to measure their distance from the expert prototypes.

$$D = \{d_{c,e} = \|z_u^i - \mathbf{ep}_{c,e}\|_2 \mid c \in \{1, \dots, C\}, \mathbf{ep}_{c,e} \in EP_c\} \quad (6)$$

where z_u^i denotes the feature of an unlabeled sample in the mini-batch. EP_c is the set of all expert prototypes belonging to class c , as shown in Eq. (3). From the distance set D , we select the top- k nearest expert prototypes set as the candidate Expert prototypes for voting.

$$D_k = \{d_{c,e} \mid (c, e) \in \text{KNN}(D, k)\} \subseteq D, \quad k = \left\lfloor \frac{1+C}{2} \right\rfloor \quad (7)$$

where KNN selects the top- k class prototypes in D with the smallest distances. The value of k is constrained by the lower bound $E_{\min}$ on the number of prototypes, as shown in Eq. (1), which prevents majority classes from dominating the voting process. C denotes the number of classes. Furthermore, distance-based weighting is adopted to assign votes with finer granularity. The distances D_k within the candidate set are first preprocessed to mitigate extreme values, resulting in a distance-based weighting term $w_{c,e}$ with a range of $(-1, 1)$, as shown in Eq. (8). Smaller values of $w_{c,e}$, including negative values, indicate that the corresponding class prototype expert is closer to the sample and therefore contributes higher voting importance. $w_{c,e}$ is then linearly mapped to $1 - \gamma \cdot w_{c,e}$, which is constrained to the range $(0, 2)$, as shown in Eq. (9).

The voting contributions from expert prototypes belonging to the same class are finally aggregated to obtain the class-wise voting score. This weighting scheme incorporates distance information while keeping the weight values within a controlled range.

$$w_{c,e} = \frac{d_{c,e} - \frac{1}{|D_k|} \sum_{d \in D_k} d}{\max_{d \in D_k} \left| d - \frac{1}{|D_k|} \sum_{d \in D_k} d \right|}, \quad d_{c,e} \in D_k \quad (8)$$

$$\text{Vote}(c) = \sum_{\substack{(c',e) \in D_k \\ c'=c}} (1 - \gamma \cdot w_{c',e}), \quad c \in \{1, \dots, C\} \quad (9)$$

where $w_{c,e}$ denotes the processed distance values obtained after centering by subtracting their mean and subsequently normalizing them. γ is a positive hyperparameter set to 0.4 that controls the range of the weights. The function $\text{Vote}(c)$ aggregates the distance-based voting contributions of the top- k expert prototypes that belong to class c , yielding the final class-wise voting score that is used to generate pseudo-labels.

$$C_{top1} = \arg \max_c \text{Vote}(c) = \tilde{y}^j, \quad (10)$$

$$C_{top2} = \arg \max_{c \neq C_{top1}} \text{Vote}(c)$$

where the class with the largest count C_{top1} is defined as the pseudo-label $\tilde{y}^j$, and the class with the second largest count C_{top2} is identified as the most confusable class.

To assess the confidence of pseudo-labels generated by expert prototypes, the module computes a vote margin, defined as the difference between the vote counts of the two most selected classes. The pseudo-label is regarded as high-confidence if its vote margin exceeds an adaptive threshold, indicating agreement among expert prototypes.

$$\text{VM} = \text{Vote}(C_{top1}) - \text{Vote}(C_{top2}) \quad (11)$$

$$M_{\text{PDEV}}^i = \mathbb{I} \left(\text{VM}_i > \frac{1}{B_u} \sum_{i=1}^{B_u} \text{VM}_i \right) \quad (12)$$

where VM_i is the vote margin of the unlabeled sample x_u^i , B_u is the batch size of unlabeled samples. C_{top1} and C_{top2} are defined as shown in Eq. (10), and the function $\text{Vote}(\cdot)$ is formulated in Eq. (9). M_{PDEV}^i denotes the selection mask for pseudo-labels generated by PDEV. Note that this selection criterion is applied to all classes. Furthermore, for pseudo-labeled samples belonging to majority classes, further downsampling is performed to mitigate the imbalance in the pseudo-label distribution.

3.2.3. Majority-class pseudo-label under-sampling strategy

For the pseudo labels that have been filtered by PDEV, further visualization analyses are conducted to examine the class distribution, as shown in Fig. 2. MPUS is designed to discard low-informative majority-class pseudo-labeled samples. The expert prototypes characterize the feature distribution of the labeled samples for each class at the current time step. We hypothesize that majority-class pseudo-labeled samples that lie farther from their corresponding expert prototype tend to exhibit lower confidence and provide more informative features for learning. Within each mini-batch, majority-class pseudo-labeled samples are evaluated by computing their distance to the nearest expert prototype.

$$d_{\min}^i = \min \left\{ \|z_{\text{major}}^i - \mathbf{ep}_{\text{major},e}\|_2 \mid \mathbf{ep}_{\text{major},e} \in EP_{\text{major}} \right\} \quad (13)$$

$$M_{\text{MUPS}}^i = \mathbb{I} \left(d_{\min}^i > \frac{1}{N} \sum_{i=1}^N d_{\min}^i \right) \quad (14)$$

where $d_{\min}^i$ is the distance to the nearest expert prototype, z_{major}^i is the feature of the majority-class pseudo-labeled sample selected by

PDEV, N is the number of majority-class samples within the batch. $\mathbf{ep}_{c,e}$ denotes the expert prototype associated with the e -th subcluster of class c , as shown in Eq. (3). The complete pseudo-label selection process is established by PDEV and MPUS to ensure the quality of pseudo-labels and to achieve a rebalanced class distribution.

$$M_{Pseudo}^i = \begin{cases} M_{PDEV}^i \wedge M_{MPUS}^i, & \text{if } \bar{y}^i \in C_{major} \\ M_{PDEV}^i, & \text{otherwise} \end{cases} \quad (15)$$

where M_{PDEV}^i denotes the PDEV mask applied over all classes, as shown in Eq. (12). M_{MPUS}^i represents the MPUS mask applied only to majority classes, as shown in Eq. (14). $\bar{y}^i$ is the pseudo-label generated by PDEV, as shown in Eq. (10).

3.3. Dual-balanced loss

Dual-Balanced Loss (DBL) is proposed to consider both minority classes at the class level and ambiguous samples at the sample level, as shown in Fig. 3. Specifically, two components are introduced: the Difficulty-Aware Weight (DAW) and the Minority-Aware Weight (MAW). A Multi-metric Extreme-value Fusion is designed to balance weights from both perspectives.

3.3.1. Minority-aware weight

To address minority classes in imbalanced data during training, Minority-Aware Weight (MAW) estimates batch-level class frequencies, assigning higher weights to classes that occur less frequently. Specifically, within each batch, the numbers of labeled and pseudo-labeled samples are counted for each class. Z-scores are then computed for classes with non-zero counts to quantify their deviation from the batch mean.

$$Freq_c = \sum_{i \in B_{labeled}} \mathbb{I}(y_i^{labeled} = c) + \sum_{i \in B_{pseudo}} \mathbb{I}(y_i^{pseudo} = c) \quad (16)$$

$$z_{mau} = \left\{ z_c = \frac{Freq_c - \mu_{Freq}}{\sigma_{Freq} + \epsilon} \mid c = 1, \dots, C \right\} \quad (17)$$

$$w_{mau}^{cur} = 1 - \gamma \cdot z_{mau} \quad (18)$$

where μ_{Freq} and σ_{Freq} denote the mean and standard deviation of the class frequencies. γ is a positive hyperparameter set to 0.4 that controls the range of the weights. w_{mau}^{cur} ensures that classes with higher frequencies receive smaller weights, while the weights remain within a reasonable range of variation. To prevent negative and extreme weight values, a unified linear offset mapping is applied within each batch, while an exponential moving average (EMA) is performed across batches. Through these operations, the weight range is effectively constrained and potential risks, such as gradient explosion, are mitigated.

$$W_{MAW} = m_w \cdot w_{mau}^{pre} + (1 - m_w) \cdot w_{mau}^{cur} \quad (19)$$

where w_{mau}^{pre} denotes the Minority-Aware Weight in the previous batch, and m_w is the EMA hyperparameter, set to 0.9, which controls the smoothing strength of the weights.

3.3.2. Difficulty-aware weight

To enhance the model's focus on ambiguous samples during training, Difficulty-Aware Weight is proposed. Within the pseudo-labeled sample subset, it leverages the relative distance between samples and expert prototypes, thereby reducing additional computational overhead. Since the expert prototypes are constructed from feature representations during training, they effectively reflect the model's current understanding of the sample features and are largely unaffected by classifier bias. By comparing the distance from each pseudo-labeled sample to the nearest expert prototype of its predicted class and the mean distance to the

prototypes of the most confusing class, the model is guided to focus on ambiguous examples.

$$d_{pred}^i = \min \left\{ \left\| \mathbf{z}_{C_{top1}}^i - \mathbf{ep}_{C_{top1},e} \right\|_2 \mid \mathbf{ep}_{C_{top1},e} \in EP_{C_{top1}} \right\} \quad (20)$$

$$d_{confused}^i = \frac{1}{|EP_{C_{top2}}|} \sum_{\mathbf{ep}_{C_{top2},e} \in EP_{C_{top2}}} \left\| \mathbf{z}_{C_{top1}}^i - \mathbf{ep}_{C_{top2},e} \right\|_2 \quad (21)$$

where $\mathbf{z}_{C_{top1}}^i$ represents the feature of a sample whose pseudo-label is predicted as C_{top1} by PDEV, and whose most confusing class is denoted as C_{top2} , as shown in Eq. (10).

DAW is applied separately to the pseudo-labeled subset and the labeled subset. For the pseudo-labeled subset, DAW measures the difficulty of each pseudo-labeled sample using a distance-based ratio and assigns it as a sample-level weight. To maintain consistency across multiple weighting metrics, $Diff_i$ is processed with the same operations as applied in $Freq_c$, namely z-score normalization. For the labeled subset, DAW adopts a focal-loss-based difficulty weighting strategy [33], where the sample difficulty is determined by the predicted probability.

$$Diff_i = \frac{d_{pred}^i}{d_{confused}^i} \quad (22)$$

$$z_{daw} = \left\{ z_i = \frac{Diff_i - \mu_{Diff}}{\sigma_{Diff} + \epsilon} \mid i = 1, \dots, B_u \right\} \quad (23)$$

$$W_{DAW}(x_i) = \begin{cases} 1 + \gamma \cdot z_{daw,i}, & x_i \in D_u \\ 1 - p_i, & x_i \in D_l \end{cases} \quad (24)$$

where μ_{Diff} and σ_{Diff} denote the mean and standard deviation of the Diff values. x_i is a training sample, D_l is the labeled data subset, and D_u is the unlabeled data subset. γ is a positive hyperparameter set to 0.4, consistent with that used in W_{MAW} . p_i denotes the predicted probability of the sample x_i obtained from the model. The distance-aware weight w_{daw}^{cur} assigns higher weights to samples that are more difficult to distinguish. Considering that pseudo labels may still contain a small number of incorrect pseudo labels, the z-score normalization and linear mapping help keep the weights within a reasonable range and prevent extremely difficult samples with incorrect pseudo labels from receiving excessively large weights.

3.3.3. Multi-metric extreme-value fusion

Traditional approaches for combining multiple indicators often rely on weighted summation or multiplication followed by averaging, which may result in information loss. When all metrics are simultaneously high or low, such fusion methods tend to generate extreme values, increasing the risk of gradient explosion. Therefore, Multi-metric Extreme-value Fusion (MEF) is designed to retain the extreme-value perspective for both majority and non-majority classes, providing fine-grained information.

For majority classes, common methods assign a small class-level weight to mitigate quantity imbalance. However, under such schemes, all samples share the same small weight, and difficult samples are not emphasized, resulting in degraded majority-class performance. MEF reduces majority-class weights at the class level while enabling attention to difficult samples within the majority class, thereby preserving majority-class classification performance. Specifically, the Minority-Aware Weight value for the majority class is used as the upper bound for the loss weights of majority-class samples. For non-majority classes, the scarcity of samples often leads to insufficient model learning. Higher weights are generally assigned to non-majority-class samples. Among these, samples that are particularly difficult to classify receive even higher weights. In the Multi-metric Extreme-value Fusion strategy, the

larger value of the two metrics is selected for non-majority-class samples. Specifically, the Minority-Aware Weight for the non-majority class is applied as the lower bound for the loss weights of non-majority-class samples.

$$W_{DBL}(x_i) = \begin{cases} \min(W_{DAW}, W_{MAW}), & x_i \in C_{\text{major}}, \\ \max(W_{DAW}, W_{MAW}), & x_i \notin C_{\text{major}} \end{cases} \quad (25)$$

$$H_{DBL}(\cdot) = -W_{DBL} \log(p_i) \quad (26)$$

where x_i is a training sample, C_{major} denotes the majority-class subset, W_{DAW} is the difficulty-based sample-level weight as shown in Eq. (24), and W_{MAW} is the frequency-based class-level weight as shown in Eq. (19). p_i denotes the predicted probability of the sample x_i obtained from the model.

Overall, under class-imbalanced conditions, the Multi-metric Extreme-value Fusion strategy explicitly classifies between majority and non-majority classes. By applying an extreme-based fusion mechanism, Multi-metric Extreme-value Fusion strategy mitigates information loss in multi-metric integration and enables dynamic, multi-perspective adaptation of loss weights.

3.4. Training pipeline

In line with most semi-supervised frameworks, PDMatch adopts a two-stage procedure consisting of pretraining and main training. In the pretraining stage, only labeled samples contribute to the supervised loss, weakly and strongly augmented views are provided as inputs and supervised using the ground-truth labels, which contain only L_s . In the main training stage, both labeled and unlabeled subsets participate. For labeled data, weakly and strongly augmented views are provided as inputs and supervised using the ground-truth labels, consistent with the pretraining stage. For unlabeled data, pseudo-labels are produced and filtered from the weakly augmented images through PDEV and MPUS, and these pseudo-labels supervise the corresponding strongly augmented images through L_u . The DBL module, responsible for assigning adaptive loss weights, is iteratively updated during training. A global algorithm summarizing the entire pipeline is provided in Algorithm 1.

Algorithm 1: The proposed PDMatch algorithm.

Input: $X = \{x_l, x_u\}$, N is the epoch number, and M is the iteration number in an epoch.

for $i = 1 : N$ **do**

 Expert prototypes init via K-means with Eqs. (1)–(3);

for $j = 1 : M$ **do**

 The candidates for voting with Eqs. (6)–(7);

 Compute the voting results with Eqs. (8)–(9);

 Generate the pseudo-label $\tilde{y}^j$ with Eq. (10);

 Filter pseudo-label samples with Eqs. (11)–(12);

if $\tilde{y}^j \in C_{\text{major}}$ **then**

 Downsample low-info pseudo-label samples with Eqs. (13)–(14);

end

 Compute W_{MAW} with Eqs. (16)–(19);

 Compute W_{DAW} with Eqs. (20)–(24);

 The extreme-value fusion of W_{DBL} with Eq. (25);

 Calculate L_s based on Eq. (28);

 Calculate L_u based on Eq. (29);

$Loss = L_s + L_u$;

 Update Expert prototypes with Eqs. (4) and (5);

end

end

$$L = \begin{cases} L_s, & \text{Pretraining} \\ L_s + L_u, & \text{Main training} \end{cases} \quad (27)$$

$$L_s = \frac{1}{2B_l} \sum_{i=1}^{B_l} \sum_{t \in \{\omega, \Omega\}} H_{DBL}(\hat{y}_l^i, p_\theta(y | t(x_l^i))) \quad (28)$$

$$L_u = \frac{1}{B_u} \sum_{i=1}^{B_u} M_{Pseudo}^i(\cdot) H_{DBL}(\tilde{y}^i, p_\theta(y | \Omega(x_u^i))) \quad (29)$$

where the labeled data are denoted as $D_l = \{(x_l^i, \hat{y}_l^i)\}_{i=1}^M$, and the unlabeled data as $D_u = \{(x_u^i)\}_{i=1}^N$. The batch consists of B_l labeled samples and B_u unlabeled samples. L_s is labeled data loss and L_u is pseudo label data loss. p_θ is the predicted probability distribution. ω is the weakly-augmented version and Ω is the strongly-augmented version. $M_{Pseudo}^i(\cdot)$ is the indicator function to filter unlabeled examples by PDEV and MPUS, as shown in Eq. (15). $\tilde{y}$ denotes the pseudo-label generated by PDEV, as shown in Eq. (10). H_{DBL} is the dual-balanced loss, as shown in Eq. (26). The supervised loss adopts pseudo-labels derived from weakly augmented images to guide the predictions of their strongly augmented versions, in accordance with the XMatch paradigm introduced in Section 2.3. This differs from the CPS paradigm [11], in which two independent encoders generate pseudo-labels for each other to perform mutual supervision. Specifically, since the model employs only a single encoder for feature extraction, pseudo-labels generated from the weakly augmented images maintain higher quality, whereas the strongly augmented branch is affected by the augmentations and struggles to produce high-quality pseudo-labels. This single-stream design requires fewer computational resources and less data, making it more suitable for medical image analysis.

4. Experiments

4.1. Datasets and evaluation metrics

To validate our framework, experiments were conducted on three datasets.

ISIC 2018: The ISIC 2018 dataset contains 10,015 images across seven categories of skin lesions and exhibits a highly imbalanced class distribution. The class distribution is shown in Fig. 1(a). The class distribution is characterized by a majority class (NV), middle classes (MEL, BKL and BCC), and minority classes (AKIEC, VASC, and DF). According to the official split [43,44], the entire dataset was randomly divided into 70% for training, 10% for validation, and 20% for testing. From the training set, we further sampled 5% and 20% as labeled subsets to establish semi-supervised learning scenarios with very limited annotations.

NCT-CRC-HE: The NCT-CRC-HE dataset [45] contains 100,000 histology images across nine colorectal cancer categories. The class distribution is shown in Fig. 1(b). The class distribution is characterized by majority classes (TUM and MUS), middle classes (LYM, DEB, BACK, ADI and STR), and minority classes (MUC and NORM). The dataset was randomly divided into 70% for training, 10% for validation, and 20% for testing. From the training set, 0.1% and 0.5% were further sampled as labeled subsets.

Fracture X-ray: The Fracture X-ray dataset is a five-category classification task comprising normal cases and four categories of humeral fractures. This dataset was privately collected from the collaborating hospital. According to the Neer classification [46], proximal humeral fractures are categorized as one-part, two-part, three-part, or four-part fractures. As shown in Fig. 1(c), the dataset exhibits a highly imbalanced class distribution, comprising 2,923/120 images for training/testing. The dataset distribution features a majority class (Normal), middle classes (One-part, Two-part, and Three-part), and a minority class (Four-part). From the training set, we extracted 20% of the data as labeled

subsets. The gold-standard labels for the fracture dataset were established through consensus among multiple senior clinicians to mitigate individual judgment bias. In addition, to allow for a comparative analysis between the model predictions and clinical assessments at different experience levels, clinicians with junior, middle, and senior expertise independently annotated the test set. It should be noted that the senior clinicians involved in the evaluation experiments were not participants in constructing the gold-standard labels.

Evaluation metrics: We employed four evaluation metrics to assess classification performance. Accuracy measures the proportion of correctly classified instances. Sensitivity reflects the model's ability to correctly identify positive cases. Specificity reflects the ability to correctly recognize negative cases. The area under the ROC curve (AUC) quantifies the model's overall discriminatory ability across varying decision thresholds.

4.2. Implementation details

All experiments were conducted on the NVIDIA RTX 4090 GPU with 24 GB of memory. Each experiment was run for 50 epochs, with the first 20 dedicated to pretraining. We used DenseNet-121 pretrained on ImageNet [47] as the backbone network. The initial learning rate was set to 0.0001. The batch size was 48, including 12 labeled samples and 36 unlabeled samples. The Adam was adopted for optimization. All input images were resized to 224×224 pixels. In line with standard practice in semi-supervised learning, various data augmentations were applied to unlabeled input images, including random horizontal and vertical flips, random rotations, and color jittering. The training, validation, and test sets were independently partitioned with no overlap. The test set was used solely to report the results of the final run. All experiments were conducted in a consistent environment to ensure fairness. Multiple random seeds were used, and the mean and standard deviation were reported. Statistical significance between experimental results was assessed using the DeLong test for AUC and the *t*-test for the other metrics.

4.3. Comparison with state-of-the-art methods

4.3.1. Evaluation on ISIC2018 dataset

Under the 20% label rate, we compared the overall performance of the proposed PDMatch with state-of-the-art semi-supervised methods on the ISIC2018 dataset, as shown in Table 1. PDMatch achieved superior results across all four major metrics. In terms of sensitivity, PDMatch achieved 79.61%, which is 3.06% higher than the next best method, ReCo [42]. This demonstrates its superior recall in identifying positive cases, particularly disease categories. PDMatch achieved the

highest AUC, Accuracy and specificity, with values of 95.32%, 93.79% and 93.19%, reflecting improvements of 0.56%, 0.01% and 0.05%, respectively. In summary, experimental results under low-label conditions validate its effectiveness and robustness.

Within the X-Match paradigm, PDMatch was further compared with other state-of-the-art X-Match-based models, focusing on the subclasses, as shown in Table 2. On the minority classes, PDMatch achieved the best performance across all metrics, with 77.96% sensitivity improved by 0.14%, with 97.01% AUC improved by 0.18%, and with 97.51% specificity improved by 0.03%. On the middle classes, PDMatch achieved the best performance across all metrics, with 76.11% sensitivity improved by 4.76%, with 93.64% AUC improved by 0.52%, and with 93.46% specificity improved by 0.09%. Although the middle classes have more samples, some of them remain hard to learn due to complex feature structures. Most state-of-the-art models underperform on these classes, likely because of an overemphasis on rare classes and neglect of middle classes. On the majority classes, PDMatch achieved the best or second-best performance across all evaluated metrics. PDMatch achieved the best AUC of 95.29% with a 0.13% improvement and the best specificity of 79.41% with a 0.08% improvement. For sensitivity, PDMatch achieved the second-best value of 95.04%, which was 0.77% lower than that of FixMatch [13]. This was largely because FixMatch [13] exhibited a notable prediction bias that increased sensitivity for the majority class while substantially degrading performance on the remaining classes. Specifically, FixMatch [13] showed a 10.68% lower sensitivity than PDMatch on the middle classes and a 4.98% lower sensitivity on the minority classes.

To further evaluate the model, experiments were conducted under extreme label scarcity with a 5% label rate, as shown in Table 1. PDMatch achieved either the best or second-best results across most evaluation metrics. PDMatch achieved the best performance on several metrics, with 71.51% sensitivity improved by 4.29%, with 91.26% AUC improved by 0.85%, and with 91.10% accuracy improved by 0.53%. For specificity, PDMatch ranked second, only 0.88% lower than the top-performing ReCo [42]. Notably, PDMatch surpassed ReCo [42] in the remaining evaluation metrics, reflecting its overall balanced and competitive performance. These results further validate its effectiveness and robustness even under conditions of severely limited supervision. The further analysis was conducted across subclasses, as shown in Table 2. PDMatch achieved the best performance across all metrics for different subclasses. In particular, for sensitivity, it reached 68.23% on the minority class with an improvement of 3.13%, and 67.83% on the middle class with an improvement of 2.52%. Besides achieving substantial improvements on the PDMatch also maintained comparatively good performance on the majority class.

Table 1

Performance comparison on the ISIC2018 dataset under different label ratios with SOTA methods. The best and second-best results are indicated in bold and underlined, respectively.

Methods	ISIC2018(20% labeled data)				ISIC2018(5% labeled data)			
	Sensitivity	AUC	Accuracy	Specificity	Sensitivity	AUC	Accuracy	Specificity
MT [5]	69.71±0.57**	92.92±0.43**	92.21±0.53*	92.34±0.34*	59.14±0.49**	86.54±0.37**	88.51±0.44**	89.62±0.24**
SRCMT [6]	71.42±0.36**	93.56±0.59**	92.59±0.88*	92.72±0.51*	61.89±0.55**	87.69±0.52**	88.68±0.52**	89.36±0.47*
FixMatch [13]	73.01±0.98**	93.91±0.32**	92.77±0.45*	91.91±0.29**	62.25±0.26**	89.87±0.43**	88.45±0.76**	88.15±0.86**
ACPL [19]	72.14**	94.36**	—	—	—	—	—	—
FreeMatch [16]	75.30±0.75**	94.64±0.51*	92.32±0.96*	92.27±0.56*	65.34±0.31**	90.35±0.59*	89.15±0.69*	89.04±0.59*
ProMatch [22]	74.79±0.42**	94.53±0.74**	92.47±0.52*	92.43±0.61*	66.31±0.53**	89.08±0.37**	88.92±0.81*	89.54±0.33*
Protocon [23]	74.45±0.65**	94.75±0.35*	93.04±0.48*	92.61±0.23*	67.15±0.72**	90.41±0.68*	90.57±0.45*	89.50±0.46*
CDMAD [35]	74.61±0.66**	94.18±0.40**	92.73±0.71*	92.74±0.72*	67.07±0.45**	84.26±0.98**	86.92±0.88**	86.64±0.75**
ReCo [42]	76.55±0.89**	—	93.25±0.61*	93.13±0.19*	64.21±0.74*	—	91.10±0.60	91.31±0.51
GTPL [17]	75.01**	94.71*	<u>93.78</u>	93.03*	60.48*	89.19**	90.03**	89.98*
CGMatch [15]	75.81±0.87**	94.76±0.59**	93.06±0.67*	93.14±0.42	67.22±0.67**	89.35±0.35**	89.11±0.47**	89.22±0.87*
PDMatch(ours)	79.61±0.79	95.32±0.45	93.79±0.58	93.19±0.35	71.51±0.59	91.26±0.40	91.10±0.56	<u>90.43±0.31</u>

¹ **: $p < 0.01$; *: $0.01 < p < 0.05$.

Table 2

Performance comparison on the ISIC2018 dataset for majority, middle and minority classes in the xmatch paradigm. The best and second-best results are indicated in bold and underlined, respectively.

Methods	Subsets ¹	ISIC2018(20% labeled data)			ISIC2018(5% labeled data)		
		Sensitivity	AUC	Specificity	Sensitivity	AUC	Specificity
FixMatch [13]	Minority	72.98±0.50**	96.12±0.36**	96.48±0.44*	51.61±0.56**	92.27±0.42*	93.86±0.73**
FreeMatch [16]	Minority	72.96±0.32**	96.16±0.35**	97.18±0.54*	57.12±0.89**	92.51±0.56*	95.65±0.55*
ProMatch [22]	Minority	75.18±0.36**	96.22±0.63*	96.78±0.57*	62.36±0.44**	91.86±0.54**	95.57±0.37*
Protocon [23]	Minority	73.12±0.51**	<u>96.83±0.57*</u>	97.43±0.39*	62.16±0.38**	<u>93.12±0.49*</u>	95.04±0.46*
CGMatch [15]	Minority	<u>77.82±0.67*</u>	96.27±0.41*	97.48±0.26*	65.10±0.66**	92.22±0.35*	95.71±0.31*
PDMatch(ours)	Minority	77.96±0.34	97.01±0.33	97.51±0.45	68.23±0.54	93.26±0.37	96.06±0.43
FixMatch [13]	Middle	65.43±0.66**	91.76±0.84**	91.71±0.47*	62.98±0.63**	86.91±0.38*	88.97±0.74*
FreeMatch [16]	Middle	71.35±0.21**	93.16±0.38**	92.27±0.39*	65.31±0.77*	87.48±0.45*	87.73±0.45**
ProMatch [22]	Middle	68.04±0.48**	92.71±0.65*	93.25±0.35*	61.62±0.39**	85.59±0.64**	<u>89.83±0.51*</u>
Protocon [23]	Middle	68.99±0.57**	92.81±0.53**	92.92±0.52*	64.12±0.41**	87.51±0.56*	89.62±0.76*
CGMatch [15]	Middle	67.44±0.43**	<u>93.12±0.39*</u>	<u>93.37±0.83</u>	61.07±0.58**	85.69±0.47**	88.64±0.31**
PDMatch (ours)	Middle	76.11±0.45	93.64±0.42	93.46±0.48	67.83±0.43	88.81±0.61	90.24±0.42
FixMatch [13]	Majority	95.81±0.94*	93.74±0.51**	78.83±0.68*	91.93±0.62**	91.51±0.47*	68.55±0.34**
FreeMatch [16]	Majority	94.19±0.64*	94.49±0.48**	77.53±0.55*	90.41±0.29*	<u>92.27±0.54*</u>	73.13±0.64*
ProMatch [22]	Majority	93.93±0.78*	94.96±0.36**	76.89±0.37**	<u>92.18±0.51*</u>	91.22±0.56*	70.61±0.71**
Protocon [23]	Majority	94.81±0.88	94.28±0.54**	77.13±0.69*	91.15±0.34*	90.92±0.78**	72.52±0.67*
CGMatch [15]	Majority	94.86±0.70*	<u>95.16±0.63*</u>	<u>79.33±0.95</u>	92.13±0.38*	91.67±0.46*	71.16±0.74**
PDMatch (ours)	Majority	<u>95.04±0.37</u>	95.29±0.51	79.41±0.53	92.36±0.66	92.58±0.35	73.98±0.58

¹ Subsets: The ISIC 2018 dataset comprises a majority class (NV), middle classes (MEL, BKL and BCC), and minority classes (AKIEC, VASC, and DF).

² **: $p < 0.01$; *: $0.01 < p < 0.05$.

Table 3

Performance comparison on the NCT-CRC-HE dataset under different label ratios in the xmatch paradigm. The best and second-best results are indicated in bold and underlined, respectively.

Methods	NCT-CRC-HE (0.1% labeled data)				NCT-CRC-HE (0.5% labeled data)			
	Sensitivity	AUC	Accuracy	Specificity	Sensitivity	AUC	Accuracy	Specificity
FixMatch [13]	85.46±0.60**	96.98±0.28**	92.67±0.62*	93.38±0.26*	93.63±0.18**	99.18±0.26*	97.13±0.17*	96.75±0.21**
FreeMatch [16]	86.17±0.81**	<u>97.17 ± 0.64*</u>	92.77±0.31*	93.42±0.27*	94.41±0.24*	99.24±0.07*	96.80±0.24*	96.93±0.27**
ProMatch [22]	85.55±0.78**	97.12±0.36**	92.82±0.43*	93.67±0.54*	<u>94.63 ± 0.30*</u>	98.87±0.11**	<u>97.25 ± 0.19*</u>	97.27±0.15**
Protocon [23]	85.14±0.72**	97.14±0.17*	92.94 ± 0.52*	93.82 ± 0.38*	94.41±0.32*	99.26±0.15**	97.14±0.64	97.55±0.33*
CGMatch [15]	<u>86.52 ± 0.58**</u>	97.13±0.49**	92.81±0.59*	93.65±0.42*	94.23±0.26**	<u>99.29 ± 0.08*</u>	96.93±0.23*	<u>97.73 ± 0.13*</u>
PDMatch(ours)	88.32±0.31	98.09±0.22	93.25±0.37	94.17±0.44	95.20±0.28	99.43±0.05	97.39±0.15	97.88±0.17

¹ **: $p < 0.01$; *: $0.01 < p < 0.05$.

4.3.2. Evaluation on NCT-CRC-HE dataset

To evaluate the performance of PDMatch under extremely low label ratios, we conducted extensive experiments on the NCT-CRC-HE dataset, which offers a larger scale and a more challenging multi-class composition. Experiments were carried out at label rates of 0.1% and 0.5%, as shown in Table 3. PDMatch significantly outperformed the other state-of-the-art models across all metrics. In particular, under the extremely scarce 0.1% label ratio, PDMatch achieved a sensitivity of 88.32%, which is 1.8% higher than that of the second-best method, CGMatch [15]. FixMatch [13] showed notably lower sensitivity and specificity, likely because data imbalance amplified classifier bias. ProMatch [22] achieved the second-best performance on several metrics under the 0.5% label ratio, as its strict filtering strategy provided reliable pseudo-label supervision. Protocon [23] obtained the second-best results on some metrics under the 0.1% label ratio, benefiting from the additional cluster-level pseudo-labels that introduced extra information. CGMatch [15] achieved the second-best performance across multiple metrics under both label ratios, as its approach of adding informative pseudo-labels is more broadly applicable.

Further comparative experiments focusing on the subclasses are provided, as shown in Table 4. PDMatch achieved the best or second-best performance across subclasses. For the minority and middle classes, PDMatch consistently attained the best performance under different label rates. In particular, under the extremely scarce 0.1% label rate, the sensitivity for the minority class reached 85.90%, which is 0.86% higher

than that of the second-best method, ProMatch [22]. The sensitivity for the middle class reached 87.79%, exceeding the second-best method, CGMatch [15], by 2.3%. For the majority class, PDMatch achieved the best performance under the 0.1% label rate and the best or the second-best under the 0.5% label rate. Although PDMatch ranked 0.1% lower than ProMatch [22] at the 0.5% label rate, it achieved superior performance on the other subclasses. Specifically, ProMatch [22] exhibited a 0.28% lower AUC than PDMatch on the middle classes and a 1.94% lower AUC on the minority classes. In summary, PDMatch demonstrated strong robustness even on more complex datasets under extremely low label ratios.

4.3.3. Evaluation on fracture X-ray dataset

Under a 20% label rate, the overall performance of the proposed PDMatch was compared with other X-Match-based models on the Fracture X-ray dataset, as shown in Table 5. PDMatch achieved the best performance among state-of-the-art models across metrics, with 69.57% in sensitivity reflecting a 0.26% improvement and 86.02% in AUC reflecting a 0.33% improvement, demonstrating the model's superior classification capability.

To validate the model's potential for practical clinical application, an additional experiment was conducted involving clinicians with different levels of expertise, including junior, middle, and senior, for comparison with human experts. The model outperformed junior and middle clinicians across all metrics, demonstrating strong overall performance.

Table 4

Performance comparison on the NCT-CRC-HE dataset for majority, middle and minority classes. The best and second-best results are indicated in bold and underlined, respectively.

Methods	Subsets ¹	NCT-CRC-HE (0.1% labeled data)			NCT-CRC-HE (0.5% labeled data)		
		Sensitivity	AUC	Specificity	Sensitivity	AUC	Specificity
FixMatch [13]	Minority	84.14±0.84*	97.13±0.70*	92.57±0.36*	95.18±0.84*	99.39±0.41*	95.93±0.51*
FreeMatch [16]	Minority	84.68±0.79*	97.44±0.56*	92.86±0.74**	96.06±0.41*	99.44±0.65*	95.12±0.68**
ProMatch [22]	Minority	85.04±0.71*	97.25±0.69*	93.10±0.49*	95.85±0.64*	97.60±0.81**	96.84±0.44*
Protocon [23]	Minority	83.12±0.41**	97.36±0.52*	93.32±0.68*	95.99±0.73**	99.37±0.69*	97.12±0.58*
CGMatch [15]	Minority	84.52±0.68*	97.19±0.54*	93.55±0.84*	96.21±0.30*	99.28±0.74*	97.27±0.39*
PDMatch (ours)	Minority	85.90±0.69	97.79±0.53	93.71±0.82	96.77±0.75	99.54±0.53	97.48±0.35
FixMatch [13]	Middle	83.95±0.32**	97.55±0.44**	96.52±0.73*	94.78±0.46*	99.25±0.39*	96.86±0.81*
FreeMatch [16]	Middle	84.74±0.53**	97.62±0.26**	96.34±0.45*	95.05±0.32*	99.31±0.25*	97.56±0.52*
ProMatch [22]	Middle	83.53±0.44**	97.67±0.39**	96.48±0.87*	95.76±0.67*	99.18±0.51*	97.47±0.66*
Protocon [23]	Middle	83.76±0.36**	97.53±0.58*	96.51±0.72*	95.82±0.55*	99.37±0.47*	97.54±0.73*
CGMatch [15]	Middle	85.49±0.32**	97.69±0.48*	96.29±0.38*	94.99±0.81*	99.40±0.73	97.80±0.38*
PDMatch (ours)	Middle	87.79±0.49	98.34±0.18	96.71±0.43	96.11±0.42	99.46±0.08	97.96±0.45
FixMatch [13]	Majority	90.59±0.79*	95.42±0.52**	86.35±0.85*	89.19±0.77*	98.79±0.42**	97.27±0.74*
FreeMatch [16]	Majority	91.23±0.74*	95.78±0.83**	86.67±0.74*	91.15±0.91*	98.85±0.39**	97.17±0.36*
ProMatch [22]	Majority	91.01±0.81**	95.61±0.78**	87.24±0.46*	90.57±0.51*	99.34±0.62*	97.23±0.62*
Protocon [23]	Majority	90.73±0.89*	95.87±0.69**	87.61±0.55*	89.30±0.72*	98.86±0.41*	98.02±0.38
CGMatch [15]	Majority	91.09±0.93**	95.66±0.84**	87.10±0.41*	90.31±0.67*	99.00±0.55*	<u>98.02±0.59</u>
PDMatch (ours)	Majority	92.05±0.85	97.75±0.41	88.30±0.59	91.34±0.83	<u>99.24±0.47</u>	98.06±0.78

¹ Subsets: The NCT-CRC-HE dataset comprises majority classes (TUM and MUS), middle classes (LYM, DEB, BACK, ADI, and STR), and minority classes (MUC and NORM).

² **: $p < 0.01$; *: $0.01 < p < 0.05$.

Table 5

Performance comparison among xmatch paradigm models and clinicians on the Fracture X-ray dataset with 20% labeled data. The best and second-best results are indicated in bold and underlined, respectively.

Methods	Sensitivity	AUC	Accuracy	Specificity
Junior Clin. ¹	44.87±0.49**	64.92±0.76**	75.83±0.46**	84.98±0.77**
Middle Clin. ¹	50.57±0.44**	68.41±0.85**	76.33±0.52**	85.53±0.58**
Senior Clin. ¹	68.23±0.57*	85.10±0.62*	84.50±0.79*	87.74±0.56
FixMatch [13]	57.06±0.95**	77.73±0.74**	82.24±0.80*	85.93±0.97*
FreeMatch [16]	69.31±0.73*	85.30±0.45*	81.75±0.46**	84.84±0.53**
ProMatch [22]	67.98±0.41*	83.61±0.87**	82.24±0.53*	86.01±0.69*
Protocon [23]	67.25±0.67**	85.55±0.69*	82.16±0.83*	85.69±0.72**
CGMatch [15]	66.23±0.63**	85.69±0.75*	82.51±0.47*	85.76±0.42**
PDMatch (ours)	69.57±0.59	86.02±0.54	<u>83.97±0.54</u>	<u>87.71±0.61</u>

¹ Independently evaluated by clinicians of the same level, and the final result is the average of their assessments.

² **: $p < 0.01$; *: $0.01 < p < 0.05$.

PDMatch surpassed senior clinicians in sensitivity and AUC by 1.34% and 0.92%, respectively. For accuracy and specificity, PDMatch was only 0.53% and 0.03% lower than senior clinicians. On the dataset with limited and imbalanced labeled samples, the model demonstrated better performance than clinicians in identifying positive cases. This capability enhances the detection of potential patients and reduces missed diagnoses. Overall, the model substantially outperformed junior clinicians and middle clinicians. It was highly effective in improving detection rates and can assist clinicians in preliminary screening to reduce missed diagnoses, with final clinicians review helping to lower false positives.

To provide a finer-grained comparison of detection rates between the models and clinicians for majority, middle and minority classes, a detailed analysis was conducted, as shown in Table 6. For the majority and middle classes, PDMatch achieved the best or second-best performance, helping to reduce missed diagnoses. For minority classes, the model performed comparably to senior clinicians. Given that clinicians routinely interpret large volumes of fracture X-rays, prolonged high workloads can increase the risk of missed diagnoses, and the model's high sensitivity and AUC can effectively alleviate this issue.

4.4. Ablation study

Extensive experiments were conducted on the ISIC2018 dataset with 20% labeled samples to evaluate the proposed model.

4.4.1. Effect of prototypes construction and hierarchical updating

In this section, ablation studies are presented on Prototypes Construction and Hierarchical Updating, as shown in Table 8. The hierarchical updating achieved the best performance across all metrics. Compared with the version using only batch-level EMA updates, sensitivity increased by 0.99%, AUC increased by 0.09%, accuracy increased by 0.53%, and specificity increased by 0.28%. Overall, the hierarchical updating has captured the latent feature distributions, which provided accurate prototype information for subsequent operations. Considering the computational cost incurred by the epoch-level update in each round, additional experiments have been provided. The theoretical complexity of one global update per epoch was estimated at 9.6249 GFLOPs. On the ISIC2018 dataset, the average runtime was 3.69 s, accounting for 6.47% of the total training time. This computational cost is acceptable because the prototype update involves only the labeled subset. In addition, different update frequencies, including 2 and 5 rounds per epoch, have been provided to meet varying resource constraints, as shown in Table 8. The version updated every epoch achieved the best performance, showing a 0.06% increase in sensitivity, a 0.04% increase in AUC, a 0.27% increase in accuracy, and a 0.15% increase in specificity compared with the version updated every two epochs. Larger gains were obtained when compared with the version updated every five epochs. In addition, the strategy employs EMA updating, with the hyperparameter m_p set to the commonly used value of 0.9, as shown in Eq. (5). Relevant experimental results are shown in Table 12.

In the Prototypes Updating stage, as shown in Eq. (4), cosine similarity is used to quantify their deep semantic relevance. This metric prevents prototype drift that would emerge if Euclidean distance were applied. Ablation experiments are provided, as shown in Table 10. The version using cosine similarity for updating and Euclidean distance for voting achieved the best performance across all metrics. Sensitivity increased by 2.57% compared with the version that updates prototypes using Euclidean distance.

For the Expert Prototype Set Construction, the feature distributions of each class were visualized, as shown in Fig. 4, revealing strong

Table 6

Performance comparison among xmatch paradigm models and clinicians on the Fracture X-ray dataset with 20% labeled data. The best and second-best results are indicated in bold and underlined, respectively.

Methods	Minority ¹			Middle ¹			Majority ¹		
	Sensitivity	AUC	Specificity	Sensitivity	AUC	Specificity	Sensitivity	AUC	Specificity
Junior. ²	58.33±0.84**	75.01±0.46**	90.66±0.58*	41.85±0.88**	60.30±0.60**	80.75±0.82**	40.47±0.76**	68.72±0.52**	91.96±0.68*
Middle. ²	61.66±0.56**	76.52±0.30**	89.91±0.62*	48.90±0.53**	64.89±0.82**	81.40±0.68**	44.48±0.58**	70.89±0.78**	91.42±0.61*
Senior. ²	83.33±0.76	94.47±0.45	91.27±0.74	59.13±0.31*	78.57±0.64**	84.82±0.74*	80.43±0.39**	95.10±0.47*	92.90±0.34*
FixMatch [13]	46.25±0.46**	72.74±0.42**	90.86±0.63*	54.51±0.74**	73.78±0.45*	82.04±0.68*	75.55±0.35**	94.55±0.81*	92.67±0.42*
FreeMatch [16]	75.62±0.72**	93.79±0.57*	90.67±0.42*	58.89±0.36*	79.27±0.41*	80.12±0.32**	86.18±0.39	94.89±0.79*	93.22±0.56*
ProMatch [22]	78.92±0.69**	93.18±0.45*	90.89±0.79*	58.37±0.61*	76.65±0.38*	81.97±0.37**	85.85±0.81*	94.86±0.52*	<u>93.23±0.79**</u>
Protocon [23]	76.66±0.71**	92.83±0.52**	91.22±0.37*	57.95±0.42*	79.48±0.56*	81.76±0.45*	85.75±0.51*	94.46±0.34*	91.91±0.87*
CGMatch [15]	67.54±0.88**	94.41±0.56*	89.79±0.48*	<u>59.19±0.68*</u>	<u>79.69±0.83*</u>	82.87±0.51*	86.02±0.73*	94.98±0.87*	90.46±0.51*
PDMatch(ours)	83.33±0.59	94.52±0.53	<u>91.26±0.65</u>	59.44±0.59	80.12±0.63	84.43±0.67	86.19±0.78	95.22±0.53	93.93±0.93

¹ The Fracture X-ray dataset comprises a majority class (Normal), middle classes (One-part, Two-part, and Three-part), and a minority class (Four-part).

² Independently evaluated by clinicians of the same level, and the final result is the average of their assessments.

³ **: $p < 0.01$; *: $0.01 < p < 0.05$.

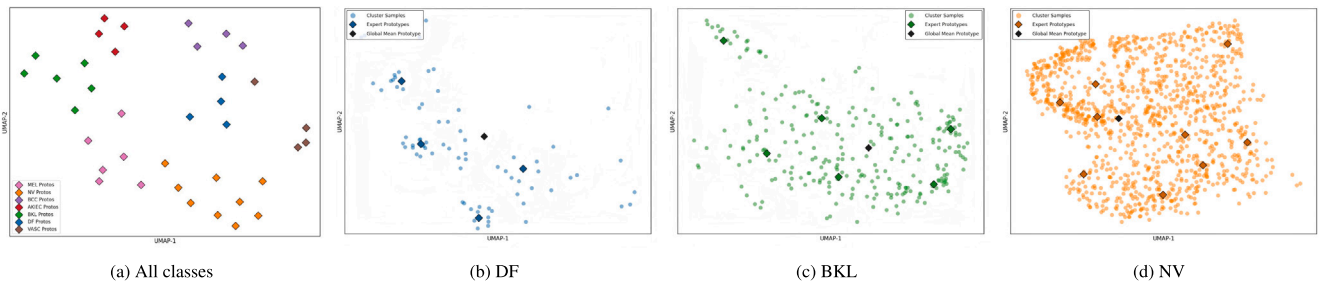


Fig. 4. The class-wise distribution of expert prototypes is visualized using UMAP. Subfigure (a) illustrates the distribution of expert prototypes across all classes, where different colors correspond to different categories. Subfigures (b)–(d) present the expert prototypes together with the labeled samples for three representative classes: the minority class DF, the middle class BKL and the majority class NV. The colors in these subfigures are consistent with those used in subfigure (a).

Table 7

Evaluation of the effectiveness of individual modules within PDMatch on the ISIC2018 dataset with 20% labeled data.

Baseline ¹	PDEV	MPUS	DBL		Metrics			
			MAW	DAW	Sensitivity	AUC	Accuracy	Specificity
✓	–	–	–	–	70.09±0.44	93.61±0.52	91.41±0.45	90.31±0.59
✓	✓	–	–	–	76.07±0.67**	95.27±0.75**	93.19±0.71**	92.13±0.37**
✓	✓	✓	–	–	76.45±0.69*	95.25±0.41	93.43±0.53*	92.46±0.33*
✓	✓	✓	–	✓	76.71±0.32	95.24 ± 0.63	93.73 ± 0.54*	92.98±0.42*
✓	✓	✓	✓	–	78.27 ± 0.57**	95.21±0.44	93.38±0.69	93.14 ± 0.23*
✓	✓	✓	✓	✓	79.61±0.79*	95.32±0.45*	93.79±0.58*	93.19±0.35*

¹ Baseline: only the supervised loss is used.

² **: $p < 0.01$; *: $0.01 < p < 0.05$. Statistical tests were conducted to compare the versions with and without the module.

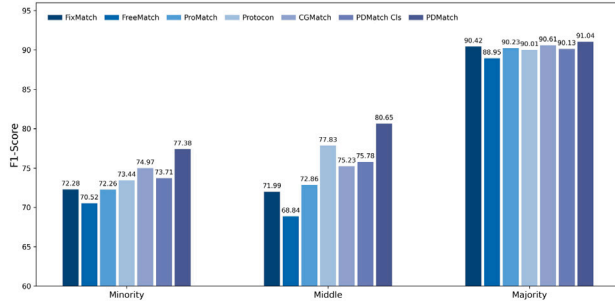
intra-class heterogeneity in the feature distribution of medical images. Traditional approaches, such as the global mean prototype, use the mean feature vector of all samples within a class as the representative. While this provides stability and accuracy, it fails to adequately capture the underlying feature distribution. The distribution of expert prototypes provides a finer-grained characterization of feature distributions, facilitating more accurate and effective pseudo-label generation and selection.

4.4.2. Effect of prototype-driven expert voting

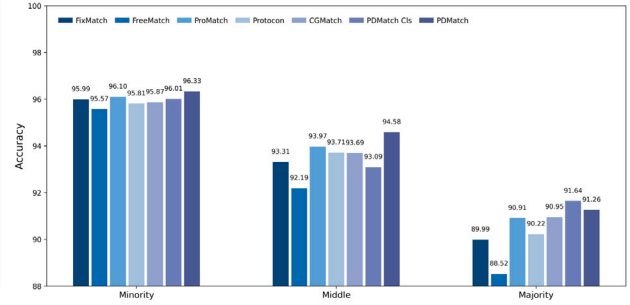
In this section, ablation studies are presented on PDEV. As shown in Table 7, compared with the Baseline, PDEV achieved notable improvements across all metrics. Sensitivity reached 76.07%, with a substantial gain of 5.98%. AUC increased to 95.27%, which is 1.66% higher than the baseline. Accuracy reached 93.19%, representing an improvement of 1.78%. Specificity increased to 92.13%, with a gain of 1.82%

over the baseline. These improvements are mainly attributed to the model learning more informative samples during the iterative training process.

Regarding pseudo-label quality, experiments were conducted to evaluate the quality of generated pseudo-labels, as shown in Fig. 5. PDMatch achieved the best results compared with state-of-the-art models across all subclasses. In terms of F1-score in Fig. 5(a), PDMatch achieved the highest performance across all subclasses. The minority classes reached an F1-score of 77.38%, which is 2.41% higher than the second-best model, CGMatch [15]. The middle classes reached an F1-score of 80.65%, outperforming the second-best model, Protocon [23], by 2.82%. The majority classes achieved an F1-score of 91.04%, exceeding CGMatch [15] by 0.43%. In terms of accuracy in Fig. 5(b), PDMatch again delivered the best results across all subclasses. The minority classes reached an accuracy of 96.33%, surpassing the second-best model, ProMatch [22], by 0.23%. The middle classes achieved an accuracy



(a) Comparison of F1-Scores across methods



(b) Comparison of Accuracy across methods

Fig. 5. Quality assessment of pseudo labels comparing PDMatch, the classifier based variants of this model, and recent x match paradigm state of the art models on the ISIC2018 dataset with 20% labeled data.

Table 8

Evaluation of the effectiveness of Prototypes Hierarchical Updating on the ISIC2018 dataset with 20% labeled data. The version without updates at both levels, where only the epoch-level initialization is used.

Batch Level	Epoch Level	Frequency ¹	Sensitivity	AUC	Accuracy	Specificity
-	-	-	78.25±0.51*	95.09±0.76*	93.00±0.83*	92.34±0.94*
✓	-	-	78.62±0.84*	95.23±0.22*	93.26±0.39*	92.91±0.33*
-	✓	1	79.23±0.48*	95.21±0.33*	93.33±0.62*	93.06±0.41*
✓	✓	5	78.63±0.65*	95.18±0.37*	93.29±0.33*	92.91±0.27*
✓	✓	2	79.55±0.76	95.28±0.26*	93.52±0.55*	93.04±0.46*
✓	✓	1	79.61±0.79	95.32±0.45	93.79±0.58	93.19±0.35

¹ Frequency: The epoch-level updating is performed every k epochs.

² **: $p < 0.01$; *: $0.01 < p < 0.05$.

Table 9

Evaluation of Adaptive Thresholding Strategies (Mean, Median, Q1, Q3, IQR) on the ISIC2018 Dataset with 20% Labeled Data.

Mehods	Sensitivity	AUC	Accuracy	Specificity
IQR	79.55±0.34	95.35±0.72	93.47±0.31*	93.08±0.26*
Q1	79.59±0.86	95.33±0.43	93.39±0.69*	93.05±0.47*
Q3	79.35±0.67*	95.31±0.67	93.74±0.60*	93.17±0.51
Median	79.54±0.42	95.37±0.85	93.78±0.29	93.12±0.39*
Mean	79.61±0.79	95.32±0.45	93.79±0.58	93.19±0.35

¹ **: $p < 0.01$; *: $0.01 < p < 0.05$.

of 94.58%, 0.61% higher than ProMatch [22]. The majority classes reached an accuracy of 91.26%, outperforming the second-best model, CGMatch [15], by 0.31%. Overall, PDMatch demonstrated consistently high pseudo-label quality, outperforming state-of-the-art models across all subclasses.

The latent space containing semantic information was used to replace the classifier in pseudo-label generation. Comparative experiments have been conducted to validate the rationale of this design, as shown in Fig. 5. PDMatch was compared with PDMatchCls by replacing the pseudo-label generation module of PDMatch with classifier outputs.

Table 10

Evaluation of Different Measures for Measuring Sample-Prototype Relations on the ISIC2018 Dataset with 20% Labeled Data.

Prototypes Updating ¹		Prototypes Voting ²		Metrics			
CosSim	Euclidean	CosSim	Euclidean	Sensitivity	AUC	Accuracy	Specificity
-	✓	✓	-	76.78±0.48**	94.95±0.75*	92.48±0.33**	92.81±0.69*
-	✓	-	✓	77.04±0.34**	95.26±0.49*	93.36±0.71*	93.14±0.18
✓	-	✓	-	78.54±0.55*	95.28±0.27	92.58±0.59*	93.07±0.27*
✓	-	-	✓	79.61±0.79	95.32±0.45	93.79±0.58	93.19±0.35

¹ Prototypes Updating: The batch-level updates, as shown in Eq. (4).

² Prototypes Voting: PDEV selects the candidates, as shown in Eq. (6).

³ **: $p < 0.01$; *: $0.01 < p < 0.05$.

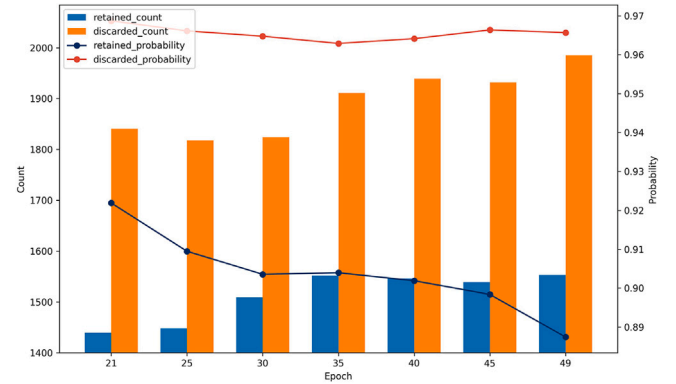


Fig. 6. Variations in the quantity and probability of selected pseudo-labeled majority-class samples within the retained and discarded subsets during training under the down-sampling strategy. The bar chart (left y-axis) shows the number of samples in each subset, while the line chart (right y-axis) represents their corresponding probabilities.

PDMatch achieved the best performance across all class subsets. For the minority classes, PDMatch improved the F1-score by 3.67% and increased accuracy by 0.32% over PDMatchCls. For the middle classes, the F1 score increased by 4.87%, and accuracy increased by 1.49%. For the majority classes, the F1-score increased by 0.91%, and the accuracy was slightly lower by 0.38%. Overall, pseudo-labels generated from encoder-based semantic features provided the highest quality.

In terms of confidence filtering, the design of PDMatch reduces the likelihood of outliers to a certain extent; therefore, the mean is adopted as a simple adaptive threshold, as shown in Eq. (12) and (14). Comparative experiments were conducted to validate this choice, as presented in Table 9. The mean achieved the best or second-best performance across metrics while also offering the lowest computational cost. Its AUC was only 0.05% lower than that of median. Q3, the strictest threshold, obtained the second-best Specificity at 93.17%, but

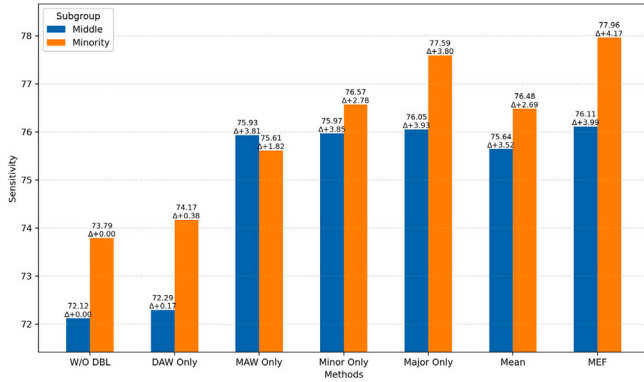


Fig. 7. Ablation experiments of DBL on middle and minority classes.

Table 11

Evaluation of the effectiveness of Multi-metric Extreme-value Fusion on the ISIC 2018 dataset using 20% labeled data.

Methods	Sensitivity	AUC	Accuracy	Specificity
W/O DBL	76.45±0.69**	95.25±0.51*	93.43±0.53*	92.46±0.33*
Mean ¹	78.71±0.62*	95.22±0.34*	93.35±0.54*	93.10±0.60
MinorOnly ²	78.88±0.58*	95.28±0.82*	93.75 ± 0.41	93.19±0.76
MajorOnly ³	79.29 ± 0.47*	95.31 ± 0.38	93.61±0.45*	93.17 ± 0.81
MEF	79.61±0.79	95.32±0.45	93.79±0.58	93.19±0.35

¹ Mean: For each sample, the mean of DAW and MAW.

² MinorOnly: Extreme-value Fusion for minority classes; simple threshold MAW for majority classes.

³ MajorOnly: Extreme-value Fusion for majority classes; simple threshold MAW for minority classes.

⁴ **: $p < 0.01$; *: $0.01 < p < 0.05$.

Table 12

Influence of hyper-parameters on the ISIC2018 dataset with 20% labeled data.

γ	w_m	w_p	Sensitivity	AUC	Accuracy	Specificity
0.2	–	–	76.31±0.41	95.01±0.81	93.20±0.65	92.57±0.78
0.3	–	–	79.09±0.78	95.45±0.47	93.61 ± 0.46	93.08 ± 0.27
0.4	0.9	0.9	79.61±0.79	95.32 ± 0.45	93.79±0.58	93.19±0.35
0.5	–	–	79.58 ± 0.96	95.13±0.59	92.24±0.39	93.01±0.41
–	0.7	–	79.07±0.45	95.31±0.61	92.89±0.44	93.13±0.56
–	0.8	–	78.86±0.61	95.33 ± 0.73	93.63 ± 0.71	93.21±0.85
0.4	0.9	0.9	79.61±0.79	95.32±0.45	93.79±0.58	<u>93.19 ± 0.35</u>
–	0.99	–	79.33 ± 0.58	95.26±0.55	93.33±0.75	93.11±0.44
–	–	0.7	79.54 ± 0.62	94.86±0.48	93.12±0.34	<u>93.15 ± 0.67</u>
–	–	0.8	79.01±0.55	95.18±0.37	<u>93.36 ± 0.67</u>	93.12±0.78
0.4	0.9	0.9	79.61±0.79	95.32±0.45	93.79±0.58	93.19±0.35
–	–	0.99	79.06±0.87	<u>95.31 ± 0.69</u>	93.42±0.61	92.85±0.61

its high threshold may discard informative pseudo-labeled samples, especially for minority classes. Q1, the most relaxed threshold, achieved the second-best Sensitivity at 79.59%, yet its overly low threshold increases the risk of introducing noisy pseudo-labels. The median and IQR, both robust to outliers, achieved the best AUC while performing slightly lower on the other metrics. Overall, the mean provides strong performance with minimal computational overhead, particularly given that the design of PDMatch already mitigates the likelihood of outliers.

In the pseudo-label selection stage, as shown in Eq. (6), Euclidean distance is used to measure their spatial separation. Spatial distance is required in the voting stage to support region-based decisions, whereas cosine similarity lacks distance information. The ablation experiments have been provided, as shown in Table 10. The version using cosine similarity for updating and Euclidean distance for voting achieved the best

performance across all metrics. Accuracy improved by 1.21% relative to the version that applies cosine similarity for voting.

4.4.3. Effect of majority-class pseudo-label under-sampling strategy

For MPUS, as shown in Table 7, incorporating this module improved sensitivity to 76.45%, representing a 0.38% increase compared to its removal. To validate the effectiveness of this strategy in removing high-confidence but low-information majority-class samples, we analyzed the mean confidence and quantity of majority-class pseudo-labels retained and discarded during model training, as shown in Fig. 6. The discarded subset generally had confidence values above 0.96 and accounted for a large proportion of samples. The retained samples had confidence values ranging from 0.89 to 0.93, indicating that PDEV, compared with probability-based models, selected more informative samples with moderate confidence, facilitating the learning of complex and difficult decision boundaries.

4.4.4. Effect of dual-balanced loss

In this section, we present ablation studies on DBL, as shown in Table 7. Given that DBL was designed based on PDEV, incorporating this module increased sensitivity to 79.61%, an improvement of 3.16%, and specificity to 93.19%, up by 0.73% compared with its removal. Further experiments were conducted to evaluate the contributions from different weighting perspectives. From the DAW alone, the model achieved an accuracy of 93.73%, an improvement of 0.3%. From MAW alone, the model achieved a sensitivity of 78.27%, an improvement of 1.82%. The dual-balanced weighting with the extremum-based fusion strategy yielded the best overall performance, confirming the effectiveness of the fusion method.

To verify the effectiveness of the Multi-metric Extreme-value Fusion weights, comparative experiments are provided, as shown in Table 11. The MinorOnly version applies the simple small weight $MAW_{majority}$ to the majority class and uses MEF only for the minority classes. The MajorOnly version applies the simple high weight $MAW_{minority}$ to the minority classes and uses MEF only for the majority class. The Mean version averages W_{MAW} and W_{DAW} at the numerical level. MEF achieved the best performance across all metrics. Compared with MinorOnly, MEF achieved a 0.73% improvement in sensitivity, a 0.04% improvement in AUC, a 0.02% improvement in accuracy, and the same best specificity, while exhibiting a 0.41% lower standard deviation. Overall, compared with various weight methods, MEF is able to exploit sample-level information to the greatest extent.

Further experiments using sensitivity for the middle and minority subsets were provided, as shown in Fig. 7. The model using only MAW achieved relatively high sensitivity, with improvements of 3.81% on middle classes and 1.82% on minority classes. By fusing DAW and MAW, MEF achieved the best overall performance, with improvements of 3.99% on middle classes and 4.17% on minority classes. This demonstrates that DAW provides additional difficulty-aware weighting and enables finer adjustment of model attention. Different fusion methods were compared, and MEF also achieved the best results. The mean-based approach showed degraded performance, 0.29% lower on middle classes than using only MAW, due to the loss of complementary information. The MajorOnly version achieved the second-best results, with improvements of 3.93% on middle classes and 3.8% on minority classes, further confirming its ability to mitigate imbalance-related bias. In addition, the strategy employs EMA updating, with the hyperparameter m_w set to the commonly used value of 0.9. Another component utilizes linear offset mapping with the hyperparameter γ set to 0.4. Relevant experimental results are shown in Table 12.

4.5. Case study

To compare the feature learning abilities of the baseline method, the recent xMatch-based model CGMatch, and the proposed PDMatch on the skin lesion dataset, Class Activation Mapping was applied for visualization and the Dice coefficient was used for quantitative analysis. For

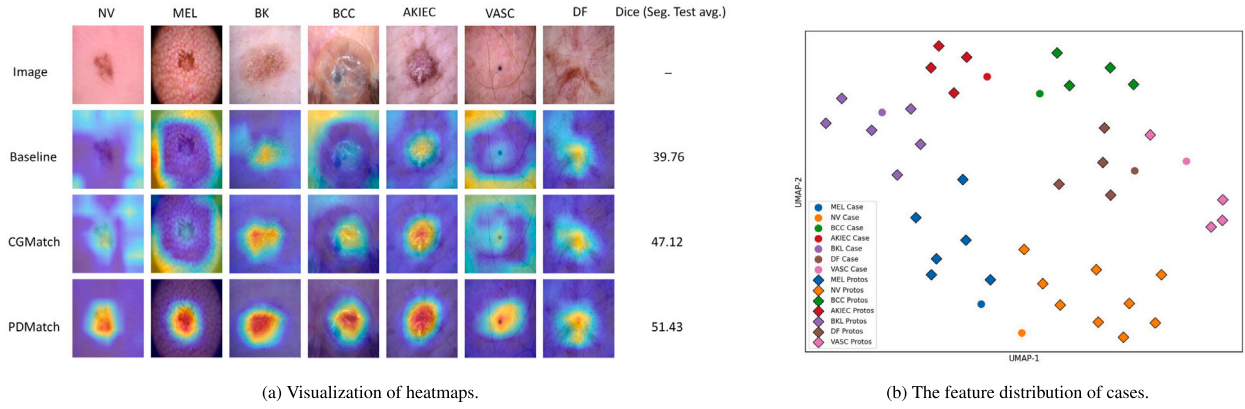


Fig. 8. (a) Visualization of heatmaps and quantitative analysis comparing PDMatch with the baseline and the recent xMatch-paradigm-based model CGMatch. (b) The feature distribution of ambiguous samples selected as cases and the distribution of expert prototypes CGMatch.

visualization, ambiguous samples were selected as representative cases, as shown in Fig. 8(b), to illustrate the ability to learn from uncertain inputs. PDMatch showed strong robustness and accurately highlighted the relevant lesion regions, as illustrated in Fig. 8(a). Because dermoscopic imaging is circular, some images such as the MEL case contain peripheral black borders, and these irrelevant artifacts were effectively suppressed by PDMatch. Hair interference in dermoscopic images, as seen in the VASC case, further challenged lesion localization for other models. For quantitative analysis, the attention distribution between lesion and non-lesion regions was assessed by computing the Dice coefficient between the heatmaps and the pixel-level gold-standard annotations, as shown in Fig. 8(a). PDMatch achieved a Dice score 4.31% higher than CGMatch, indicating stronger feature learning capability. It should be noted that because only sample-level labels were used during training, the Dice metric in this study evaluates the heatmaps rather than serving as a segmentation-performance measure.

4.6. Sensitivity analysis of hyper-parameters

The hyperparameters were determined through one-factor-at-a-time sensitivity analysis, as shown in Table 12. γ is a positive hyperparameter (set to 0.4) that controls the range of the weights in the linear mappings, ensuring that the assigned weights remain within an appropriate interval, as shown in Eq. (9), (18), and (24). $m_p = 0.9$ is a hyperparameter that controls the update ratio of EMA, as shown in Eq. (5). m_w is the EMA hyperparameter, set to 0.9, which controls the smoothing strength of MAW, as shown in Eq. (19).

5. Discussion

5.1. Hierarchical prototype updating

In terms of the prototype updating strategy, the hierarchical updating is proposed, in which epoch-level clustering is performed at each epoch, followed by batch-level EMA updates. Although the prototypes have been updated via EMA, the batch-level EMA update relies on the fixed subclusters from the initial clustering, which may accumulate global biases as training progresses. The clustering at each epoch enables the subclusters to be updated according to the latest feature distributions. The additional ablation experiments have been provided, as shown in Table 8. The hierarchical updating achieved the best performance across all metrics. Several computational cost analyses are provided in Section 4.4.1. The overhead introduced by the epoch-level updating remains within an acceptable range and leads to a clear improvement in model performance.

Moreover, cosine similarity is used in the batch-level updating process, as shown in Eq. (4). Cosine similarity effectively measures the deep semantic relevance between sample features and class prototypes,

which is primarily reflected in the feature directions. This measure prevents prototype drift that would otherwise occur if Euclidean distance were applied, as demonstrated in Table 10. The version that uses cosine similarity to update prototypes achieved the best performance, highlighting the superiority of cosine similarity in class-prototype construction.

5.2. Pseudo-label generation and selection

In terms of pseudo-label generation, the latent space produced by the encoder is used to replace the classifier outputs, considering that the latter carries limited information and is susceptible to systematic bias [48]. Although similar biases may still exist in the encoder, the latent space contains substantially richer semantic information. The decoupling strategy is based on this principle and has demonstrated favorable performance on imbalanced data [49]. Therefore, the latent space with semantic representations was employed to replace the classifier for pseudo-label generation. The comparative experiments were conducted to validate the rationale of this design, as shown in Fig. 5. Overall, pseudo-labels generated from encoder-based semantic features provided higher quality than those produced by the classifier, particularly for non-majority classes.

In terms of pseudo-labeling voting in PDEV, the different number of class prototypes is designed to accommodate data imbalance, while potential bias may arise during the voting stage. PDMatch has been designed to minimize the bias. The expert prototypes of each class have been shown to form relatively concentrated spatial distributions rather than being randomly scattered across the feature space, as shown in Fig. 4. The candidate prototypes participating in voting have also been filtered using distance-based KNN. Therefore, the influence of majority-class absolute prototype counts may arise when the distances to multiple candidate classes are similar. Considering that the number of class prototypes is generated by the Effective Number of Samples, as shown in Eq. (1), the absolute differences remain relatively comparable. Overall, a potential bias related to prototype counts may exist, and the design of PDMatch is intended to minimize this risk.

MPUS is an under-sampling operation applied to majority-class pseudo-labels. The samples selected during down-sampling are drawn from the high-confidence subset of majority-class pseudo-labels, as shown in Fig. 4, which exhibit very high confidence. Therefore, MPUS does not increase the risk of introducing incorrect pseudo-labels, and its effectiveness in improving model performance has been validated through ablation experiments, as shown in Table 7.

Moreover, Euclidean distance is used during the pseudo-label selection stage to measure the spatial separation between sample features and class prototypes. Spatial distance is required in the voting stage to support region-based decisions, whereas cosine similarity lacks explicit distance information, as demonstrated in Table 10. The version

that employs Euclidean distance for pseudo-label voting achieves the best performance.

5.3. Dual-balanced loss weighting

DBL is a dual-perspective loss weighting method that assigns larger weights to non-majority classes and smaller weights to majority classes, as shown in Eq. (25). In addition to improving the performance of non-majority classes, which are the primary focus of the task, this design is also necessary for maintaining majority-class performance. Majority-class samples share the same class-level weight MAW based on class frequency, and each sample has an independent difficulty weight DAW based on the difficulty of pseudo-label generation. For the majority-class weighting strategy, the small weight MAW for high-frequency majority classes is used as the upper bound on DAW, ensuring that the final weights remain smaller than MAW, as shown in Eq. (25). This design has been adopted to reduce majority-class weights at the class level by using MAW as the upper bound, while enabling attention to difficult samples within the majority class, thereby preserving majority-class classification performance. If only a small class-level weight were applied across samples, difficult samples would not be emphasized, resulting in degraded majority-class performance. The additional comparative experiments have been conducted, as shown in the new Table 11. Overall, compared with various weighting methods, PDMatch is able to exploit sample-level information to the greatest extent.

In the MEF strategy, difficult pseudo-label samples belonging to non-majority classes receive higher weights. The potential risk does exist when incorrect labels receive increased weights. Therefore, PDMatch has been designed to minimize this risk from two directions. First, the pseudo-label subset has relatively high quality, as shown in Fig. 5, allowing DAW to assign higher weights to correct and informative samples, thereby enabling the model to learn important information. Second, for samples that may contain potential errors, the weights were uniformly constrained through preprocessing such as Z-scores and linear mapping, as shown in Eq. (17) and (23). These operations prevented extreme weights from being assigned to incorrect samples. Further experimental analysis showed that incorrect pseudo-labels accounted for 16.88% of all samples whose weights were increased by DAW. The effectiveness of DAW was validated in the ablation study, as shown in Table 7. Overall, although a potential risk does exist, it has been effectively controlled, and DAW has shown an overall positive influence on the model.

5.4. Future work and limitations

Overall, PDMatch proposes PDEV and MPUS from the perspective of pseudo-label generation and selection, and proposes DBL from the perspective of loss weighting. The framework demonstrates superior performance on three datasets with complex and imbalanced feature structures. The current work does not include an explicit correction algorithm for addressing the encoder's classification bias, which remains an important research direction. Several studies [38,50] have investigated strategies that further reduce the bias in the encoder through contrastive learning and adversarial generation. These strategies will be further examined in the next stage of our studies. In addition, on the private Fracture X-ray dataset, the data volume will be further expanded, and the development of pixel-level annotations will be advanced to support future research.

6. Conclusion

In this work, we investigate semi-supervised medical image classification under complex data challenges, including class imbalance, label scarcity, and intra-class heterogeneity. The proposed PDMatch framework comprises Prototype-Driven Expert Voting (PDEV), Majority-class Pseudo-label Under-Sampling (MPUS), and Dual-Balanced Loss (DBL). PDEV is a non-probability-based selection strategy that enables adaptation to complex data distributions. This prototype-guided approach for generating and selecting pseudo-labels leverages latent information in

the feature space to obtain high-confidence pseudo-labels. MPUS serves as a concise post-processing strategy built upon PDEV, providing rebalanced sampling specifically for pseudo-labels. This helps mitigate prediction bias that may be exacerbated by imbalanced pseudo-label data. The under-sampling concept can be further extended to other pseudo-labeling frameworks to broadly alleviate imbalance within pseudo-label subsets. DBL offers fine-grained weighting, jointly attending to minority classes and difficult samples by incorporating both the difficulty signals from PDEV and class-frequency information into the loss training. Extensive experiments are conducted on three datasets with distinct imbalance structures, verifying the effectiveness of PDMatch in addressing these data challenges and demonstrating its practical value. Although prototype construction and hierarchical updating introduce some computational overhead, PDMatch remains advantageous and practical in scenarios involving complex feature structures and low label rates.

CRedit authorship contribution statement

Sulei Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Shihao Sheng:** Writing – review & editing, Visualization, Formal analysis, Data curation, Conceptualization. **Tiegong Wang:** Writing – review & editing, Visualization, Validation, Formal analysis, Data curation. **Xudong Liang:** Validation, Data curation, Conceptualization. **Xiao Chen:** Validation, Data curation. **Hao Shen:** Visualization, Validation, Data curation. **Haoyang Zhang:** Validation, Investigation. **Jiang Xie:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Guangchao Wang:** Visualization, Validation, Supervision, Funding acquisition, Data curation. **Jiacan Su:** Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Guangchao Wang reports that article publishing charges were provided by the Key R&D Program of the China Ministry of Science and Technology. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

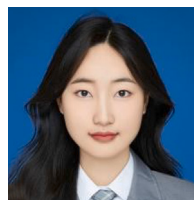
The ISIC 2018 dataset and NCT-CRC-HE dataset used in this paper are publicly available. The Fracture X-ray dataset is not publicly available due to patient privacy and ethical restrictions.

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Author biography



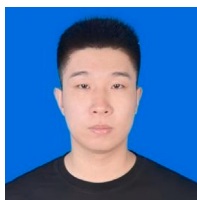
Sulei Wang received the B.E. degree in Computer Engineering and Science from Jiangsu University, Zhenjiang, China in 2023. She is currently pursuing the M.E. degree in Computer Engineering and Science in Shanghai University. Her research focuses on semi-supervised learning for medical images with complex features.



Shihao Sheng received the B.M. degree in Clinical Medicine from the Naval Medical University, Shanghai. He is currently a resident physician in the Orthopedics Department of Xinhua Hospital affiliated with Shanghai Jiao Tong University School of Medicine. His research focuses on trauma treatment and limb salvage for diabetic foot.



Tiegong Wang received his Ph.D. in Imaging Medicine and Nuclear Medicine from the Naval Medical University. He is currently an Associate Chief Physician in the Department of Radiology and Diagnostic Imaging at the First Affiliated Hospital of the Naval Medical University. His main research focuses on the imaging diagnosis of musculoskeletal and abdominal diseases.



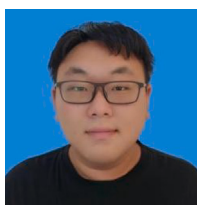
Xudong Liang received the B.E. degree in Computer Engineering and Science from Southwest Forestry University, Kunming, China in 2022. He is currently pursuing the M.E. degree in Computer Engineering and Science in Shanghai University. His research focuses on knowledge distillation and transfer learning for medical multimodal models.



Xiao Chen received his Ph.D. in Clinical Medicine from the Second Military Medical University and completed postdoctoral training in Materials Science at Fudan University. He currently serves as an Associate Chief Physician, Associate Professor, and Master's Supervisor in the Orthopedics Department at Xinhua Hospital Affiliated with Shanghai Jiao Tong University School of Medicine. His research and clinical work focus on the diagnosis and treatment of knee joint disorders, including complex trauma, joint replacement, and arthroscopic surgery.



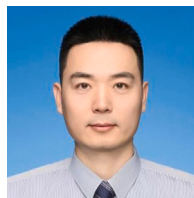
Hao Shen received his Ph.D. in Clinical Medicine from Shanghai Medical College, Fudan University. He currently serves as an Associate Chief Physician in the Orthopedics Department at Xinhua Hospital Affiliated with Shanghai Jiao Tong University School of Medicine. His clinical expertise focuses on complex intra-articular fractures, minimally invasive fixation, and joint injuries of the shoulder, elbow, and knee.



Haoyang Zhang received the B.E. degree in Network Engineering from Henan Normal University, Xinxiang, China in 2023. He is pursuing a Ph.D. in Computer Engineering and Science in Shanghai University. His research focuses on applying deep learning methods to medical imaging and spatial omics data processing.



Jiang Xie obtained a Ph.D. in computer application technology from Shanghai University in 2008. From September 2011 to December 2012, she was a visiting associate researcher in the Department of Mathematics at the University of California, Irvine, USA. She is a professor at the School of Computer Engineering and Science at Shanghai University. She has been working on knowledge discovery from biomedical big data, with her main research directions including the study of the molecular mechanisms of complex diseases, AI-based disease risk analysis, and detection and recognition in medical imaging. Across these areas, her work has involved algorithm design, model development, and the development of applied software systems. She is currently a senior member of the Chinese Computer Federation (CCF) and a member of the Technical Committee of Open System and Parallel Computing under the CCF.



Guangchao Wang received his Ph.D. in Clinical Medicine from the Second Military Medical University and currently serves as an Associate Chief Physician in the Orthopedics Department at Xinhua Hospital Affiliated with Shanghai Jiao Tong University School of Medicine. His research interests include applying artificial intelligence to orthopedic trauma care and minimally invasive fracture treatment.



Jiacan Su received his Ph.D. in Clinical Medicine from the Second Military Medical University and completed postdoctoral training in Materials Science at East China University of Science and Technology. He is currently a Changjiang Distinguished Professor and serves as Director and Chief Physician of the Orthopedics Department at Xinhua Hospital affiliated with Shanghai Jiao Tong University School of Medicine. His research focuses on osteoporotic fractures, bone aging, and bone organoid transformation.